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Do Investments in Clean Technologies Reduce Production Costs?

Insights from the Literature

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Abstract

In response to growing environmental concerns, particularly climate change, governments have encouraged innovation and adoption of clean technologies through various policy measures. At present more than half a trillion US dollars is being invested annually in clean technologies. Based on the existing literature, this study analyzes whether investments in clean technologies increase productivity. The findings are mixed. Employing firm-level data, the majority of ex-post studies show a positive relationship between clean investments and firms' productivity, especially in the energy-intensive manufacturing sector. Most studies for the transport, building and power sector use an ex-ante, technology or sectoral level analysis instead of ex-post analysis to examine the economics of clean technologies. In the transport sector, transportation services with electricity or hydrogen are still more expensive than that with gasoline and diesel vehicles. Some studies, however, project that cleaner vehicles will be economically attractive within a decade. Many studies report that clean technologies reduce energy consumption and save energy bills in the building sector, although some studies do not agree. Most studies for the power sector indicate that renewable technologies have not yet reduced the average costs of grid electricity because of their intermittency and a smaller share in the total electricity supply.

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Do Investments in Clean Technologies Reduce Production Costs? Insights from the Literature[1]

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Do Investments in Clean or Green Technologies Reduce Production Costs? Insights from the Literature

1. Introduction

In response to growing concerns to address climate change, one of the biggest challenges to human development, development financing institutions including national governments, international financial institutions, bi-lateral donors, non-governmental organizations, and the private sector have channeled hundreds of billions of dollars towards green or clean or low carbon economic development. The International Energy Agency (IEA) estimates that about US$600 billion was invested in clean energy technologies in 2019, of which US$343 in renewable energy production technologies and US$249 billion in energy-efficient technologies were invested (IEA, 2020a). The total investment in clean energy in 2019 accounted for 34% of the global investments in entire energy technologies, including fossil fuel technologies. The multilateral development banks (MDBs) allocated, on average, US$51.2 billion annually for low carbon development over the last five years (EBRD, 2020). It is likely that most development institutions, including the World Bank Group, would prioritize clean or low carbon technologies in their assistance for infrastructure development in developing countries. However, a question emerges – how long would such supports be needed for clean and low-carbon technologies before the market takes them on its own? The market is expected to voluntarily invest in clean technologies in the absence of government policies if such investments are profitable. These investments are profitable if they reduce the production costs of firms where energy is a significant input. Is there any evidence that a deployment of clean technologies has contributed to lowering production costs of various energy-consuming sectors, such as industries, power utilities, transportation, buildings, and agriculture?

This study aims to answer this question by digging into existing literature. Existing studies that use ex-post empirical techniques (i.e., econometric analysis) as well as ex-ante economic analysis or technological/sectoral modeling are reviewed. The ex-post studies examine the relationships between investments in green or clean technologies and production costs or productivity of firms using observed data from the field. Whereas, the ex-ante analyses examine the economic or financial viability of a clean technology using current or projected information on technological and economic variables (e.g., energy efficiency and costs of technologies, fuel prices). The former technique is more common at the firm level analysis. The latter is widely used at the technology level. The ex-post analysis can directly relate investment on clean technologies with production costs of firm or productivity. The ex-ante analysis focusses on whether investments on clean technologies reduce costs of energy service delivery. For example, if investments in green technologies or renewable energy technologies (e.g., solar, wind, hydro, geothermal) have helped reduce the average electricity supply costs. Similarly, in the transportation sector whether investments in cleaner vehicles, such as electric, hydrogen, and hybrid vehicles, reduce the cost of transportation services. Finally, in the buildings sector (residential or commercial), if the adoption of energy-efficient appliances or devices causes net economic gains to the households or business owners.

The study carries out an extensive review of existing empirical literature to understand the relationship between green/clean technologies and production costs in general and energy service costs in particular sectors: manufacturing, transportation, electricity, buildings, and agriculture. While retrieving existing studies, multiple databases and numerous keywords are used. The databases include EconLit, Scopus, Web of Science, Google Scholar, JSTOR, and organizational databases such as International Energy Agency and the World Bank Group. It includes peer-reviewed journal articles, and academic literature and documents produced by international organizations, research institutions, and government departments. The keywords used are 'clean energy', 'clean energy investment', 'clean energy and productivity', 'low carbon technologies', 'economics of clean energy', 'green energy', 'green energy technologies', 'modern energy', 'energy technology and economy', R&D in energy technologies', 'energy and total factor productivity', 'clean energy and total factor productivity'.

The findings of the existing studies are mixed and inconclusive. Most studies examining the relationships between green/clean technologies and productivity show a positive relationship. However, the channels through which the investments are translated into productivity gains are not clear. Studies at the sectoral level (e.g., electricity and transportation) do not provide clear evidence of cost reductions in transportation services and electricity supply due to renewable energy technologies and electric or hydrogen vehicles. A similar outcome is observed in the buildings sector. However, investments in energy-efficient technologies, particularly in energy-intensive industries reduce production costs or improve productivity.

The remainder of this paper is structured as follows: Section 2 presents the trends of green/clean investment. Section 3 discusses the findings of the existing studies that examine the relationship between the investments in green/clean technologies and productivity in general. It also highlights some factors influencing the relationship. In Section 4, a green/clean investment effects on production costs in specific sectors, such as manufacturing, power, buildings, and transportation are analyzed. This is followed by the discussions of other factors that drive private sector investments in green/clean technologies in Section 5. Finally, key conclusions are drawn in Section 6.

2. Clean Technologies – Investment and Expansion

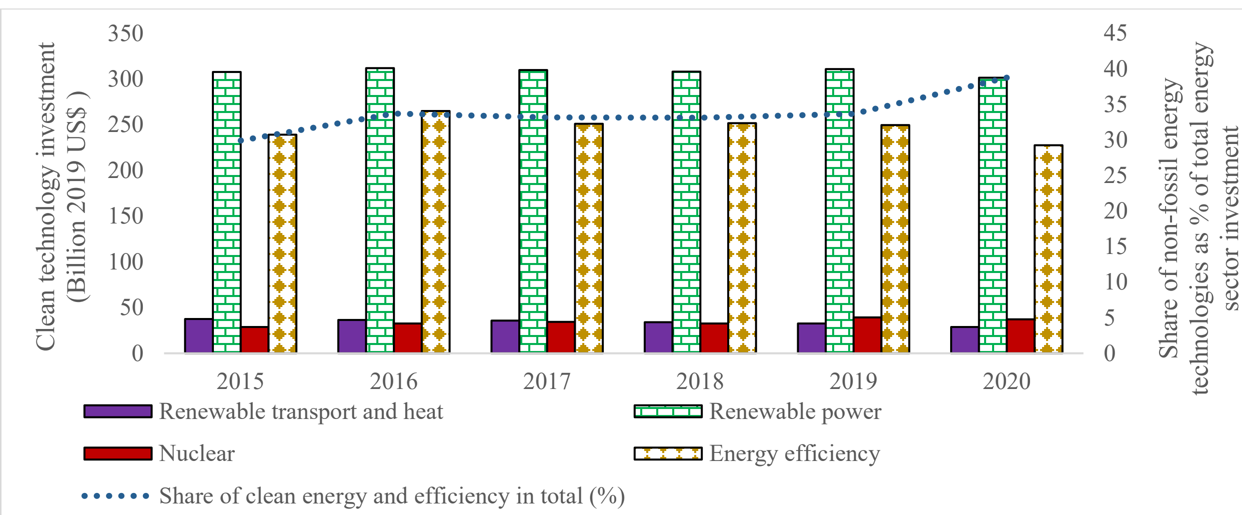
Over the last two decades, clean energy technologies, including both in the energy supply side and energy demand side, have attracted a large investment each year. Government policies to promote green/clean energy technologies to address global climate change and local air pollution problems are the main drivers of these investments. This section briefly highlights the investment trends and their results (e.g., expansion of renewable energy capacity).

Investments in clean technologies, particularly renewable electricity generation technologies, have rapidly increased since 2000. The IEA estimates that about US$308 billion was invested in renewable electricity technologies in 2015 (Figure 1). It increased to US$311 billion in 2019. Investments in renewable fuels for transportation (i.e., liquid biofuels) have, however, decreased from US$38 billion in 2015 to US$33 billion in 2019. These investments are also reflected in Figure 2 that portrays the expansion of renewable electricity generation capacities vis-a-vis fossil-fuel-based capacities since 2000. Yet, the global share of electricity generation from renewable technologies in total electricity generation accounted for about 10% if hydropower is excluded. Including hydropower, the share of renewable electricity generation in total electricity generation stands at 28% in 2020 (Figure 2a). Including nuclear, non-fossil fuel-based technologies produced 38% of the total electricity globally in 2020. In terms of capacity to generate electricity, the shares of renewables, renewables including hydro, and non-fossil fuels technologies are 20%, 37%, and 43%, respectively (Figure 2b). The reason for the lower share of generation from renewables compared to that of capacity is that renewables electricity plants (e.g., solar, wind, hydro) can operate fewer hours in a year than fossil fuels-based plants.

Although the share of renewables is relatively small in total installed capacity, they are the largest in terms of new capacity addition. For example, from 2010 to 2020, the capacity addition of solar power is the largest (675 GW) followed by the wind power (552 GW) (Figure 2c). Solar and wind together accounted for almost half of the total capacity added during the 2010-2020 period. Due to the stringent climate change targets, particularly if the net-zero path is followed, the share of renewables (solar, wind, hydro, and other renewable energy technologies (RETs)) would be much higher by 2030.

Investments in renewables are expected to accelerate due to the current debate on the net-zero target (i.e., a goal to have net-zero GHG emissions globally) by the middle of this century. The net-zero target is consistent with the 1.5 °C targets -- limiting the increase in the global mean surface temperature of the earth's surface to 1.5 °C above the pre-industrial level--. To stay in the path of meeting this target, the IEA estimates that the world needs 630 GW of solar PV and 390 GW of wind to be added annually by 2030 (IEA, 2021a). It also projects that two-thirds of the total energy supply should be met from wind, solar, bioenergy, geothermal, and hydro energy by 2050 to realize the net-zero target.

Figure 1. Investments (Billion US$) in non-fossil fuel-based technologies and their share (%) in total energy investment (2015-2020)

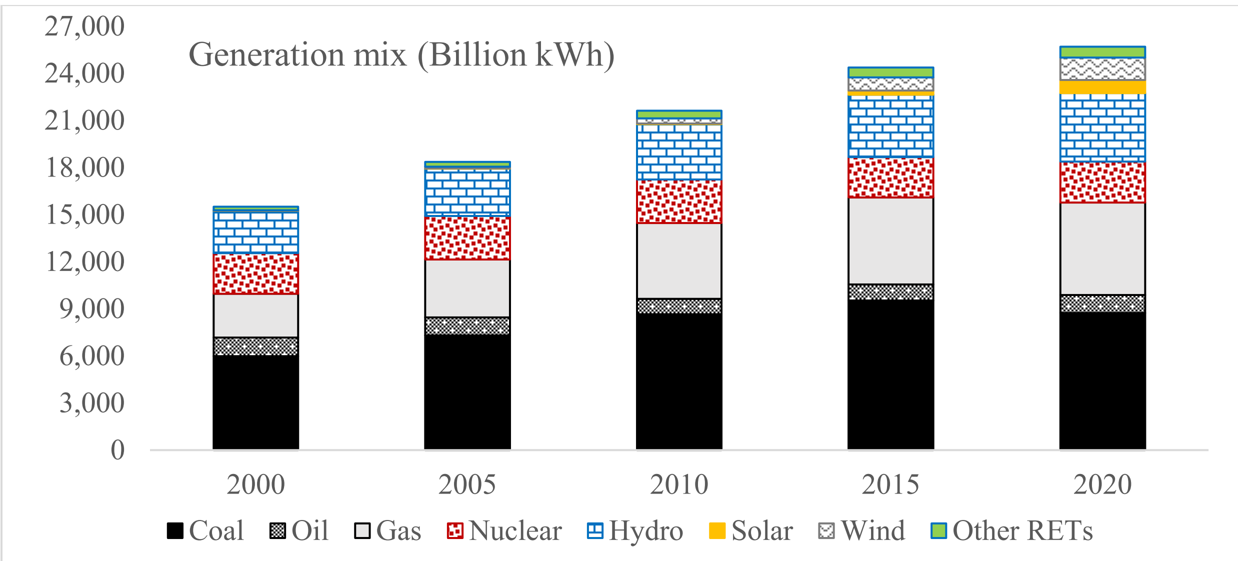


Source: IEA (2020a)

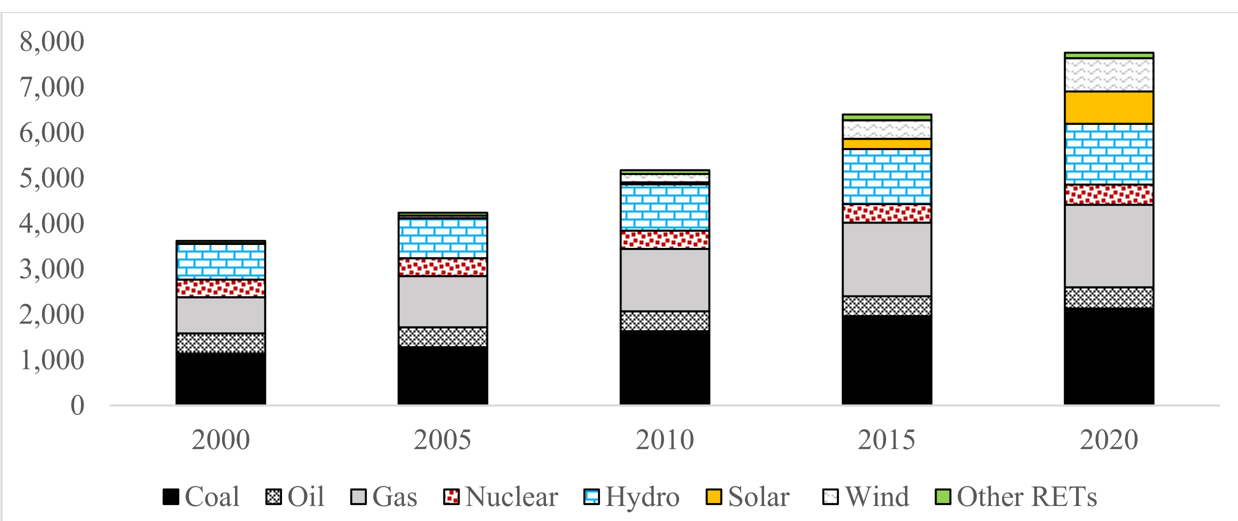
There is a sharp rising trend in the number of corporations joining the 100% renewable energy consumption (RE100) group, which brings together organizations that have set a target to source 100% of their power from renewables by a particular date in the future. The number of corporate members of RE100 increased from 12 in September 2014 to 222 in January 2020 (FS et al., 2020). The RE100 includes prominent companies such as Apple, Facebook, and Microsoft. The list includes 19 of the world's 100 largest companies by revenue.

Figure 2. Global electricity generation and capacity mix

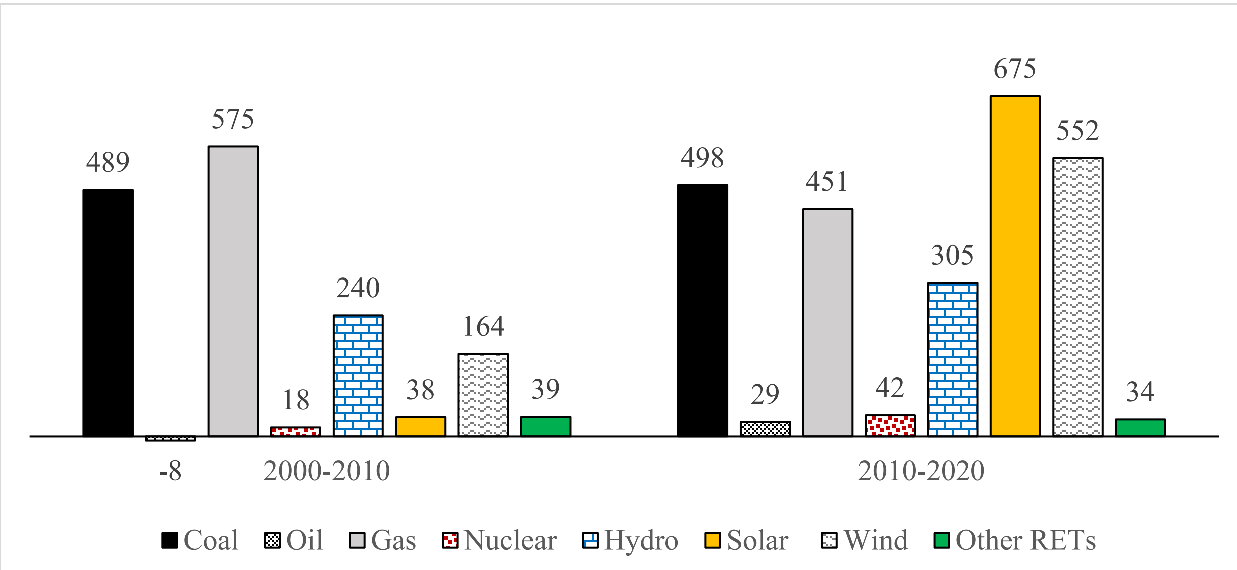
(a) Annual generation mix (Billion kilowatt-hours)



(b) Total installed capacity mix (GW)



(c) New capacity or capacity added during 2000-2010 and 2010-2020 period (GW)



Other RETs refer to other renewables, including biomass, geothermal, tidal, etc.

Source: IEA online database. https://www.iea.org/data-and-statistics/

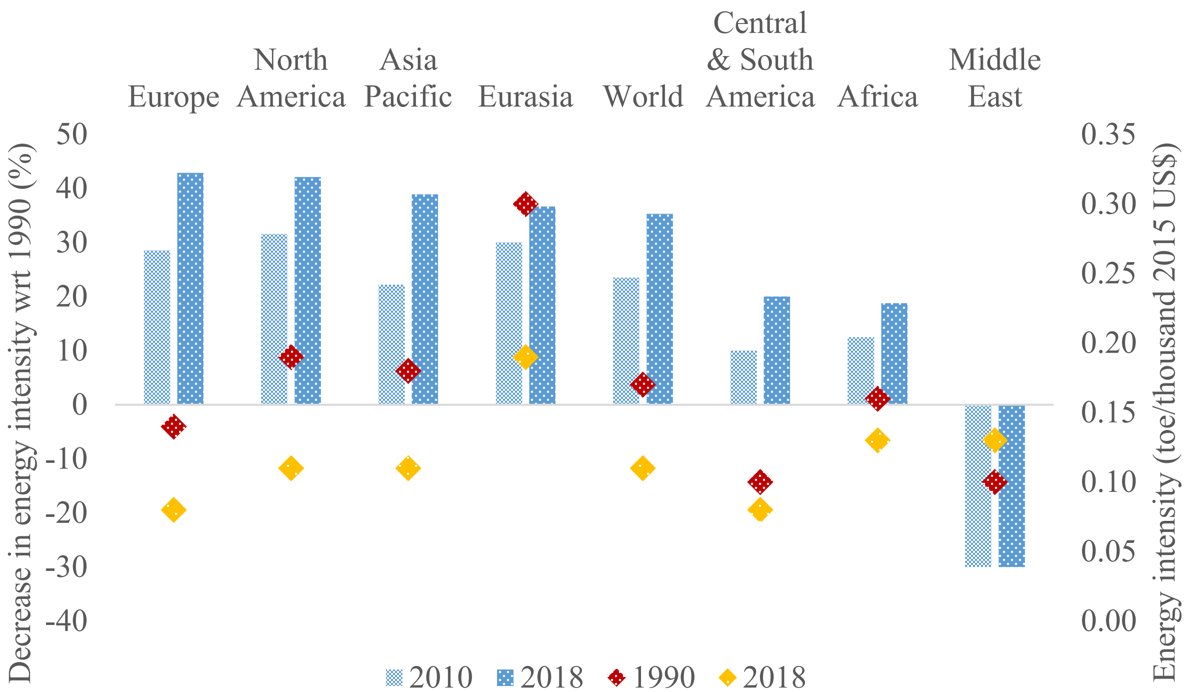
The total investment in clean technologies, including renewables and energy efficiency (EE), has remained stagnant at around US$600 billion during the 2015-2019 period. It dropped to US$562 billion in 2020 due to the COVID pandemic (Figure 1). One key reason for the downward trend of investment despite continued expansion of clean technology adoption is falling unit costs for key renewable technologies (e.g., solar PV, wind, and electric vehicles).

The EE improvement means using technology that requires less energy to perform the same task. It has multiple benefits, including reducing energy bills[3], mitigating climate change, reducing air pollution, improving energy security, and increasing energy access (IEA, 2020b). The IEA, (2020c) report estimates that the efficiency gains since 2000 in IEA member countries resulted in

[3] In some cases, EE improvements might not reduce energy consumption due to rebound effect.

the avoidance of over 15% or US$600 billion more energy expenditure for fuels for heating, road transport, and a wide range of other energy end-uses.

Figure 3. Energy intensity (EI) in 2010 and 2018 relative to 1990 (bar, left axis) and EI in 1990 and 2018 (scatter plot, right axis) by world regions



Source: IEA (2021b)

The improvement of EE is one of main driver of reducing in energy intensity (EI) in many countries and regions around the world.[4] For instance, excluding the Middle East, the worldwide EI of the economy – the amount of energy used to generate a unit of GDP – has decreased in the past, ranging from a low of 19% in Africa to a high of 43% in Europe between 1990 and 2018 (Figure 3 bar graph). The increasing trend in EI in the Middle East is mainly due to the dominance of expanding energy-intensive industries, commodity exporting-based economies, and low energy prices. In absolute values, the EI is the highest in Eurasia (0.19 toe per thousand 2015 US$) to the

[4] Lower (or higher) EI could indicate that energy is being used efficiently (or inefficiently) but not always.

lowest in Europe and Central and South America (0.08 toe per thousand 2015 US$) (Figure 3 scatter plot).

3. Green/Clean Investments and Firm's Productivity in General[5]

A large number of existing studies investigate the relationship between investments in cleaner or greener technologies and the cost of production or productivity. In general, productivity measures the efficiency with which firms, organizations, industry, and the economy convert inputs (e.g., labor, capital, and raw materials) into output. Productivity grows when output grows faster than inputs when both inputs and outputs are measured at the same or constant prices. Depending upon the input in consideration, there are many different measures for productivity, such as single-factor productivity (e.g., labor productivity or capital productivity) or multifactor (e.g., capital and labor or capital-labor-energy-materials) productivity. It can be measured at the firm or industry, or organization or country level (OECD, 2001). Unless specified otherwise, productivity in this study refers to total factor productivity (TFP)[6]. In this section, the literature that presents overall relationships between green/clean energy investment and productivity at the industry level are discussed, followed by the section on various attributes, such as firm size, ownership, and regulations, that can influence the relationship.

3.1. Relationship between green/clean investment and productivity

[5] We have divided the discussions on clean/green investments and their impacts on production costs in two sections. The studies analyzing the relationship using the firm level data are presented in Section 3. The firms could be from any sector, manufacturing, electricity supply, transportation services. In Section 4, we discuss studies that focus on a particular sector such as a manufacturing sector (e.g., chemicals), electricity generation sector, power sector, building sector and agriculture sector.

[6] The TFP, also known as multi-factor productivity, is a measure of the output of a firm or industry or economy relative to the size of all its primary factor inputs. It measures the residual growth that cannot be explained by the rate of change in the services of labor, capital and intermediate outputs of firms or industries or economy as whole. It is often interpreted as the contribution to productivity growth made by factors such as new knowledge, technologies, and experience.

Existing studies argue and demonstrate empirically that stricter environmental regulations cause the firms to invest in innovation and adoption of cleaner technologies, which would eventually lead to increased productivity. There is a strong argument that environmental regulations cause technological innovation that will increase productivity and that the productivity gain offsets the costs of environmental regulations (Porter, 1991; Porter and van de Linde, 1995). This hypothesis, popularly known as 'Porter's hypothesis', has been tested empirically by many studies that are discussed in this paper. The empirical results are, however, ambivalent. While earlier studies do not necessarily prove Porter's hypothesis, more recent studies do. For example, Jaffe and Palmer (1997) investigate whether the stricter environmental regulations in the United States have spurred R&D activities (i.e., increased patents of green/clean technologies) using panel data of manufacturing industries from 1975 to 1991. They find investments in pollution control technologies (i.e., clean technologies) have a significant positive effect on R&D expenditures. However, they do not find strong evidence of the deployment of clean technologies during that period. This is sometimes referred to as a "weak" version of Porter's hypothesis in the literature because it does not confirm whether innovation or the use of clean/green technologies is good for firms. The result is not surprising, though, as the large-scale deployment of cleaner or greener technologies in local air pollution control occurred after the 1990s. Moreover, increased deployment of greener energy technologies, such as wind and solar, accelerated more recently, after 2005.

Some studies (e.g., Cao et al., 2021; Peng et al., 2021) also examine impacts on the productivity of green/clean investment resulted from environmental regulations in China. Cao et al. (2021) report that regulated firms are more motivated (or obligated) to invest in green/clean technologies to meet existing environmental regulations. Using sample data from environmentally dirty industrial firms (e.g., coal mining and washing, oil and gas mining, petroleum processing, coking and nuclear fuel processing, power, thermal production and supply, and gas production and supply) for the 2000-2007 period, they find that investments in green/clean technologies led to an increase firms' productivity. Using firm-level panel data of Chinese industrial enterprises for the 1998-2007 period, Peng et al. (2021) provide the evidence of a positive relationship between the deployment of new green/clean technologies resulted from flexible environmental regulations and productivity of the firms.

Several studies directly examine whether an investment in green/clean technologies increase firms' productivity or production costs (Hamamoto, 2006; Lanoie et al., 2008; Böhringer et al., 2012; Horvathova, 2012; Rubashkina et al., 2015; Tugcu and Tiwari, 2016; Palmer and Truong, 2017; Rath et al., 2019; Leoncini et al., 2019; Stucki, 2019; Sohag et al., 2021). Table 1 summarizes the methods used and the main findings and arguments of selected studies. Based on Japanese manufacturing industries, Hamamoto (2006) finds that the pollution control expenditures have a positive relationship with R&D expenditures and have a negative relationship with the average age of capital stock. It also shows that increases in R&D investment stimulated by the regulatory stringency have a significant positive effect on the TFP growth. Lanoie et al. (2008) find that investments in pollution-control technologies, as a proxy of stringent environmental regulations, resulted in modest, long-term gains in TFP in a sample of 17 Quebec manufacturing sectors. They show that this effect is more important in industries that are more exposed to international competition. Using a panel dataset of German manufacturing sectors for the 1996-2002 period, Böhringer et al. (2012) examine the relationship between environmental policies, including green/clean investment, and firms' productivity growth. They do not find a positive impact of environmental investment on production growth. They, however, conclude that environmental policies or regulations should stimulate environmental investment to realize firms' productivity growth. Analyzing 79 global firms between 2007 and 2012, Palmer and Truong (2017) find a positive relationship between new green technologies use and firms' profitability.

Using data from 25 OECD countries, Sohag et al. (2021) examine if the investment in renewable energy technologies promotes TFP. They find that a higher share of renewable energy in the production process significantly promotes TFP in the long run. This is not necessarily the case in the short run. The increased human capital, innovations, and trade openness augment the positive relationship between renewable energy investment and TFP. However, the paper does not explain the channels through which increased renewable energy improves TFP. Similar effects of renewable energy on TFP are also reported by Rath et al. (2019) when they examine the relationship between the growth of renewable energy and TFP using panel data of 36 countries for the 1981–2013 period. They show that an increased TFP is associated with an increased consumption of renewable energy, whereas the consumption of fossil fuels is negatively associated with TFP. The findings are, however, vary across different regions. Since increased consumption of renewable energy reflects increased investment in renewable energy technologies, the findings of the study imply that green investments increase TFP at the aggregated level. Using a dataset of 5,498 manufacturing firms in Italy from 2000 to 2008, Leoncini et al. (2019) show the positive effect of green technologies on firm's productivity growth is greater than that of non-green technologies.

There exists no consensus, however, on the relationship between the investment of green/clean technologies and productivity. Using the firm-level data from the Czech Republic, Horvathova (2012) finds a significant upfront innovation expenditure that led to firms' lower productivity. However, this negative effect also tends to fade over time and transforms into a positive effect in the long run. Using cross-country sector-level panel data for 17 European countries between 1997 and 2009, Rubashkina et al. (2015) find no evidence of a positive relationship between pollution abatement and control expenditures, as a proxy for clean/green technology investment, and productivity. Analyzing the causal relationship between the growth of TFP and that of fossil and renewable energy consumption in the power sector in BRICS countries[7] during the 1992-2012 period, Tugcu and Tiwari (2016) find no remarkable causal link between renewable energy consumption and TFP growth. Instead, they find a positive relationship between non-renewable energy consumption and TFP in Brazil and South Africa.[8] Using firm-level data of selected firms collected through a survey in Austria, Germany, and Switzerland, Stucki (2019) finds significantly positive productivity effects of an investment in green energy technologies in firms with high energy costs but not in firms with medium energy costs. It finds significantly negative effects for firms with low energy costs.

3.2 Factors affecting the relationships between clean investments and productivity

Several factors influence whether investments in green/clean technologies increase a firm's productivity. These factors include the size of firms, their ownership, the presence of environmental regulation, and activities for investments (development vs adoption of technologies). Below, we briefly highlight these factors.

As discussed above, there are mixed findings on the relationships between green/clean investments and productivity. Firm size is one of the key factors that influence whether green/clean

[7] BRICS refers to the group of countries including Brazil, Russia, India, China and South Africa.
[8] Note, however, that expansion of renewable energy is happening more recently. Therefore, this finding is not surprising as the data used in this study were for the period when there was little expansion of renewable energy.

investment impacts a firm's productivity. The reason is simple. Large firms have a higher investment capacity, especially in the case of energy-intensive firms. Therefore, they tend to invest in green/clean technologies. On the other hand, small firms, particularly less energy-intensive ones, may not find it beneficial, as suggested by some literature (e.g., Stucki, 2019). Hrovatin et al. (2016) argue that large firms which are energy-intensive and exposed to international competition are more likely to invest in energy-efficient technologies. Large firms also have financial resources for investment in clean or energy-efficient technologies. Chen et al. (2021) find small-scale firms in China are notably less likely to invest in the development (or innovation) of clean technologies (measured in terms of patents) because of budget constraints. On the other hand, large-scale firms can invest in innovation (R&D) because of their access to finance and innovation risk diversification capabilities. With environmental regulations, small firms choose to reduce their output rather than increase investment in technologies to meet the regulations (Chen et al. 2021).

Qi et al. (2021) elaborate on why the firm size is an important factor to influence the productivity effects of green investment, presenting a case of investment in water pollution control technologies in China. They find that large firms can install clean technologies without impacting their outputs as the fixed costs of clean technologies are relatively small. On the other hand, small firms that do not invest in clean technologies face higher costs due to different regulatory costs. They also argue that the fixed costs associated with the installation of clean technology led to increasing returns to scale above a threshold size of a firm (i.e., large firms). That is the reason why large firms are less pollution-intensive than small firms in China.

Does the ownership of firms influence the productivity effect of green/clean investment? Not enough empirical evidence is available to answer this question. However, large-scale multinational firms usually have private ownership (i.e., multinational companies). As explained above, such firms invest in green/clean technologies if they are energy-intensive and exposed to international competition. They normally do not face financial constraints. Similarly, large state-owned companies, such as electricity utilities, have financial resources for green/clean investment (e.g., adoption of renewable technologies for power generation). Therefore, the size of the company appears to matter more than the ownership regarding investment in green/clean technologies. Yet, some studies attempt to differentiate the impacts of green/clean investment based on ownership. Cao et al. (2021) and Peng et al. (2021), for example, find the impacts on the productivity of green investments to meet the existing regulatory policies (i.e., cleaner production standard) are greater for state-owned firms than that for non-state-owned firms in China. This finding, again, boils down to the size of the firms because state-owned firms in China account for a significant proportion of large-scale energy firms, and cleaner production standards have greater effects on the improvement of productivity of state-owned firms.

3.3 Effects of innovation vs. adoption of green/clean technology on the productivity

Whether the firm's productivity change due to green investment is caused by the investment in the deployment of green/clean technologies or by their development (i.e., R&D investment)? While the increased adoption of green/clean technologies directly affects productivity even in the short run, innovation of new technology through R&D investment does so in the long run. However, the development of new technology does not guarantee its adoption. The discussions earlier are related mostly to investment in the adoption or deployment of clean technologies. In general, the countries with greater energy R&D investment intensity show higher levels of investment in the green technologies (Popp et al., 2011). Here, we briefly highlight the literature that exploits the relationship between R&D investment in clean technologies and firms' productivity. Some existing studies empirically show that environmental knowledge spillover or the knowledge created through R&D investment on environmental or clean technologies are positively related to firms' performance.

Based on a survey of small and medium-sized Italian manufacturing firms over the 2004-2006 period, Aldieri et al. (2021) show a positive but only slightly significant effect of process or product innovation on productivity. They also show positive and strongly significant effects if it is an organizational innovation.[9] Using firm-level data on inputs, outputs, and their patents registered from Japan, the USA, and European countries, Aldieri et al. (2020) show investment in environmental or clean technologies or knowledge, measured in terms of registered patents, affect firms' productivity significantly and positively. Using firm-level panel data of Dutch manufacturing firms over the 2000-2008 period, van Leeuwen and Mohnen (2017) find a positive

[9] Generally, product innovation includes new products, process innovation includes new technologies and organizational innovation includes new business practices (OECD, 2005).

relationship between innovation (investment in clean/green technology R&D) and productivity (TFP).

However, in some cases, the green/clean technology innovations hurt firms' productivity, mainly due to additional direct costs associated with R&D investments to innovate and modify production design and process, and increasing financial constraints due to the fixed budget for investment in innovation activities (Zhang et al., 2020; Zhang et al., 2019). Synthesizing a review of 65 different studies on the evidence of a relationship between R&D investment, both green/clean and other R&D investments, and firm or industry productivity in OECD countries, Ugur et al. (2016) report a weak and highly heterogeneous relationship. They find the sources of heterogeneous are the size of the firms (small or large), funding sources of R&D (private or public), and the magnitude of R&D investment (low or high).

Table 1. Selected empirical studies examining the relationship between green/clean investment and productivity

Study	Method	Main findings and argument
Qi et al. (2021)	An econometric method is applied to 39 Chinese manufacturing industries.	Firm size plays a key role in clean technology adoption. Environmental regulations eliminate some firms with low production efficiency through market competition and increase the overall productivity level.
Sohag et al. (2021)	CS–ARDL technique is used in 25 OECD countries from 1980 to 2015.	A higher share of renewable energy in the energy mix in the production process significantly promotes TFP in the long run, while it is inconclusive in the short run.
Cao et al. (2021)	PSM-DID method is applied to Chinese energy firms from 2000 to 2007.	The cleaner production regulations increased regulated firms' TFP, especially small firms. Also, the impact of cleaner production regulations of state-owned firms was greater than that of non-state-owned firms.
Peng et al. (2021)	A DID method is used to a panel data of Chinese industrial enterprises for the 1998-2007 period.	The market-based environmental regulation exerted significant productivity-enhancing effects across all types of industrial enterprises, with stronger effects associated with privately owned, more productive, and less pollution-intensive enterprises.
Aldieri et al. (2020)	The empirical method using panel data on firms' patents on environmental innovation from 9 European countries, Japan, and the USA for the 2002-2007 period.	Green technology innovation and its knowledge spillovers affect firms' TFP significantly and positively in all the investigated economic areas.
Leoncini et al. (2019)	A quantile regression method is used for 5498 manufacturing firms in Italy from 2000 to 2008.	There is a positive effect of green technologies on a firm's growth. A firm's experience appears important for the growth benefits of green technologies.
Stucki (2019)	The ordinary least square model is applied using firm-level data to Austria,	Firms with relatively high energy costs show significantly larger marginal effects of green technology investments on TFP than do firms with low

	Germany, and Switzerland in 2014.	energy costs. However, this effect is observed only for 19% of firms with the highest energy costs.
Rath et al. (2019)	The panel cointegration tests and dynamic OLS are applied to panel data of 36 countries for 1981-2013.	Generally, the increase in renewable energy consumption positively affects the TFP growth in the long run, while the increase in fossil fuel consumption has a mixed effect on TFP growth.
Zhang et al. (2019)	The modified fixed investment model is applied to data of 34000 Chinese firms in 2013.	The environmental abatement crowds out innovation investment and decreases firms' productivity due to financial constraints.
Palmer and Truong (2017)	Data on 1020 technological green products between 2007 and 2012 by 79 global firms are used.	The relationship between technological green new product introductions and firm profitability is positive.
Hrovatin et al. (2016)	The probit and bivariate probit models are applied to the panel data set of 848 Slovenian manufacturing firms during 2005-2011.	Large firms with high energy costs tend to invest both in EE and green technology to increase productivity and to improve their competitive position in the market. Non-environmental investments by the firms do not crowd out green/clean technology investments.
Tugcu & Tiwari (2016)	A panel bootstrap Granger causality analysis of Konya is used for BRICS countries for 1992 to 2012.	The relationship between green technology and TFP growth is less substantial than that between non-green technology and TFP growth. In some cases, green technology has a positive effect on TFP growth.
Hottenrott et al. (2016)	A two-stage least squares regression analysis of firm-level panel data of Germany covering the period from 2000 to 2008.	Firms can achieve higher TFP gains from adopting new technologies if they adapt their organizational structures. Such complementarity effects may be of particular importance for the adoption of GHG abatement technologies.
Rubashkina et al. (2015)	The instrumental variable estimation approach is used for manufacturing sectors of 17 European countries between 1997 and 2009.	There is a positive impact of environmental regulation on the output of innovation activity, as proxied by patents, thus providing support in favor of the "weak" Porter hypothesis.
Horvathova (2012)	A measure of environmental performance is applied to the firm-level data from the Czech Republic.	The effect of environmental performance on financial performance is negative in the short run but it is positive in the long run.
Böhringer et al. (2012)	Econometric analysis is used for a panel dataset of German manufacturing sectors.	The environmental regulations stimulate environmental investment to be compatible with the pursuit of production growth.
Lanoie et al. (2008)	An econometric approach is used for 17 sectors in the Quebec manufacturing industry for 1985-1994.	The impact of environmental regulation on productivity is negative in the short run but it is positive in the long run. This effect is stronger in a subgroup of industries that are more exposed to international competition.
Hamamoto (2006)	A reduced-form econometric model is used for 5 Japanese manufacturing industries in the 1960s and 1970s.	The pollution control expenditures have a positive relationship with the R&D expenditures and a negative relationship with the average age of capital stock. Also, increases in R&D investment have a significant positive effect on the growth rate of TFP.

4. Green/Clean Investment at the Sectoral Level

Earlier, we discussed the productivity effects of green/clean investment based on ex-post studies that consider firms from multiple sectors. In this section, we dive deep into green/clean investment effects on production costs in specific sectors, such as manufacturing, power, buildings, and transportation. The indicators used for productivity here differ across sectors and they are not necessarily the TFP considered earlier. For the manufacturing sectors, we still use the TFP as the measure of productivity. Where TFP data is not available or not relevant, we use the cost of energy services because manufacturing sectors, particularly the energy-intensive ones, invest in energy-efficient technologies to reduce their production costs. Private firms invest in energy-efficient devices or processes if the investment is profitable (or it saves overall production costs). Therefore, a reduction in energy costs can be interpreted as a reduction in production costs. For the electricity sector, we look at whether an expansion of renewables sources of electricity generation (hydro, solar, wind) reduces the overall costs of electricity supply from a grid. In the transportation sector, the indicator is cost reduction for transportation services due to clean or alternative vehicles. For the buildings sector, for example, we look at whether the EE measures reduce households' or businesses' utility (energy or electricity) bills. However, studies examining the impacts of the adoption of clean energy technologies are limited in the agriculture sector.

4.1 Manufacturing sector (excluding transportation and electricity generation sectors)

Energy is an important input to manufacturing industries. Costs of fuels and electricity constitute a major share in the total cost of production in energy-intensive industries, such as iron and steel, aluminum, cement and glass, pulp and paper, chemicals, and fertilizers. These industries often invest in energy-saving technologies to reduce their energy bills. Some countries have regulations that mandate improvements in EE in industrial establishments. In such a case, Porter's hypothesis would hold if the investments in EE improvements reduce overall production costs or improve productivity.

Several recent studies provide empirical evidence to support the positive relationship between improvements in EE and productivity (Li et al., 2021a; Huang and Wu, 2021; Haider & Bhat, 2020; Filippini et al., 2020; Unver & Kara, 2019; Arriola-Medellín et al., 2019; Zuberi & Patel, 2019; Xylia et al. 2017; Morrow III et al. 2013; Hasanbeigi et al. 2012). Using detailed firm-level survey data collected through the Chinese Annual Industrial Survey for the 2003-2008 period and using multiple empirical strategies and identifications, Filippini et al. (2020) estimate the impact of a national EE program on firms' TFP growth in China's iron and steel industry. They find a positive and statistically significant effect. Using a simulation model to analyze the impacts of adopting selected energy-efficient technology in the chemical industry, Li et al. (2021a) find that adopting clean technologies would reduce more than 40% of the production costs for the same output level from the industry. Huang and Wu (2021) report a reduction in production costs in the cement industry due to energy-efficient technologies in Taiwan. The results of several studies are presented in Table 2. Most studies show positive impacts of clean or energy-efficient technologies on productivity. As expected, the impacts vary across the studies. Some studies, such as Boyd and Pang (2000), explain how an increased EE enhances the productivity of a firm, presenting an example of the glass industry, which is an energy-intensive industry.

Some studies find that investments in the development or deployment of clean technologies also increase labor productivity. Using data from firms involved in developing and adopting energy-efficient technologies during the 2012-2014 period in Austria, Germany, Switzerland, Arvanitis et al. (2017) find that the investments on clean technologies increase labor productivity. However, the positive relationship between the adoption of clean technologies and a firm's productivity could be conditional on other policies or activities. For example, Hottenrott et al. (2016) find that firms gain productivity by adopting environmental technologies only in the presence of supportive organizational structures. Using the panel data on the adoption of environmental technologies in German manufacturing and service firms for the 2000-2008 period, they find firms that adopted green technologies together with changes to their organizational structures utilize green technologies better than those firms that only adopted green technologies without having supportive organizational structures.

Table 2. Selected empirical studies examining the relationship between clean investment and cost of production in the manufacturing sector

Study	Method	Findings
Li et al. (2021a)	Simulation method using Aspen Plus software for the chemical industry.	Retrofitting the coal chemical industry (CCI) with more efficient petrochemical technologies reduced the total annual costs (operating and capital expenditure) of CCI by 41.3% with total product yield almost unchanged.

Huang & Wu (2021)	Simulation methods using extended energy conservation supply and extended MAC curves are applied to the cement industry in Taiwan.	Using energy-efficient technologies, by 2025, the total annual fuel savings could reach 1828 TJ, equivalent to 6% of the energy used for cement production in 2018.
Zhang et al.(2021)	A network Epsilon Based Measure model and Tobit regression are applied to the Chinese construction industry for 2000–2017.	In the construction industry, market-friendly environmental regulations have a positive effect on green technology innovation, while rigid and mandatory environmental regulations (e.g., command and control) hurt green technology innovation.
Wan et al. (2021)	A generalized method of moments regressions is applied to Chinese energy firms for 8 weeks before and after the pandemic.	Although the impact of COVID-19 on firms' financial performance is negative for both types of firms (clean energy and fossil-fuel firms), the investor attention to the disruptive effects of the pandemic had a significant and positive effect on clean energy stocks' returns.
Haider & Bhat (2020)	An econometric technique is used in the paper industry in 21 states in India from 2001-2013.	An increasing level of TFP is associated with the lower level of EI or better EE. Better skills and capacity utilization have a positive impact on EE performance of the paper industry, while the heterogeneity within the structure of the industry hurts EE performance.
Filippini et al. (2020)	The two-stage approach of translog cost function and DID approach are applied to 5340 Chinee iron & steel firms for 2003 and 2008.	The national EE program had a positive and statistically significant effect on productivity in iron & steel firms. The EE program increased annual productivity growth by 3.1% points, with approximately equal contributions from technical change and scale efficiency change.
Palčič & Prester (2020)	A four-step OLS is applied to survey data collected from 232 Croatian and Slovenian manufacturing firms during 2018-2019.	Green innovation has an impact on firms' competitive advantage through their image, and not necessarily through financial gains. Advanced manufacturing technologies also contribute to both the firm's productivity and green innovation.
Gong et al. (2020)	An OLS method is applied to high-pollution firms (Shanghai and Shenzhen) in China from 2009 to 2018.	The influence of rising labor costs on green technology innovation has a threshold effect. The effects of the rise of labor costs on the green technology innovation of high-pollution firms illustrate an "inversely U-shaped" variation trend with the increase of the degree of market monopoly.
Haas et al. (2020)	The least-cost optimization model is applied for the electricity supply of copper mines in Chile, Peru, China, the USA, Austria, Indonesia, and Mexico from 2020 to 2050.	At present, it is attractive for the copper mines to have a solar generation of 25% to 50% of the yearly electricity demand. In the future, the expected electricity costs range from 60 to100 Euro per MWh for 2020 and from 30 to 55 Euro per MWh for 2050, with the lower bound in sunny regions such as Chile and Peru.
Zuberi & Patel (2019)	EE cost curves are applied to the Swiss chemical and pharmaceutical industry from 2000 to 2012.	Electricity savings by improving motor systems is estimated at 15% of the total electricity demand in 2016. The EE potential also varies with different energy prices.
Unver & Kara (2019)	A regression analysis using AMPL Software is applied to a steel forging facility in Turkey.	Correct regulation of the production process through increased EE would result in 65% energy savings in the unit production of a steel forging facility. Considering the total annual production of the facility, 7,639 kWh of

		energy may be saved in a year, which is equivalent to €3,686 per year or 6.6 ton of oil equivalent per year.
Arriola-Medellín et al. (2019)	An energy management system is applied to the oil and gas processing center in Mexico.	Investment in EE at the oil and gas processing center reduced the consumption of natural gas and electricity consumption by 75% and 98%, respectively.
Shapiro & Walker (2018)	Statistical decomposition is applied to the manufacturing industry in the United States from 1990 to 2008.	Overall, between 1990 and 2008, air pollution emissions from US manufacturing fell by 60% despite a substantial increase in manufacturing output mainly due to changes in environmental regulation (pollution tax) rather than changes in productivity or the composition of products produced (trade).
Zhu et al. (2018)	A global data envelopment analysis (DEA) is applied to the mining& quarrying industry in China from 1991 to 2014.	The TFP of China's mining & quarrying industry increased by 71.7%. Technological progress was the most important contributor, and the decline in scale efficiency and management efficiency were the two inhibitors. However, the green TFP growth and the returns to scale of the sub-sectors within the industry are mixed.
Wurlod & Noailly (2018)	A trans-log cost function and decomposition analysis are applied to a set of 14 industrial sectors in 17 OECD countries for 1975-2005.	Green innovation contributed to the decline in EI in most sectors. The median elasticity of EI to green patenting is estimated at −0.03, i.e., a 1% increase in green patenting activities in a given sector is associated with a 0.03% decline in EI. Overall, half of the decrease in EI is related to changes in input prices and a half to changes in production technologies.
Xylia et al. (2017)	A cost-benefit analysis is applied to Swedish energy-intensive industries.	The benefit-to-cost ratio of the impact of implementing EE obligation scheme for Swedish energy-intensive industries ranges from 1.56 to 2.17 and the break-even cost ranges from 83.3 to 86.9 Euro per MWh. The estimated annual energy savings potential is 1.25 TWh per year.
Morrow III et al. (2013)	A bottom-up energy conservation supply curves model is applied to the Indian cement industry for 2010-2030.	The cumulative cost-effective plant-level electricity savings potential for the Indian cement industry for 2010-2030 is estimated to be 83 TWh, and the cumulative plant-level technical electricity saving potential is 89 TWh during the same period.

4.2 Transportation sector

It is not straightforward to measure productivity in the transportation sector because of the mixed ownership of vehicles between households and businesses. Automobiles (e.g., cars, SUVs, motorcycles, small trucks) are usually owned by private households, whereas bigger vehicles (e.g., busses, big trucks) are owned by private companies or government utilities. In this study, we use the costs of providing transportation services as an indicator to compare the economics of road vehicles. It measures the unit costs of delivering comparable transportation services.[10] The costs are the life-cycle costs that include costs of owning and operating the vehicles (Comello et al., 2021). This measurement is commonly used in the literature. It is also referred to as 'total cost of ownership (TCO)' to compare the overall cost of clean and alternative vehicle technologies (e.g., BEVs, PHEVs, and FCEVs) with that of conventional vehicles (i.e., ICEVs).[11] With advances in technologies, the TCO of clean vehicles has been falling significantly, recently. For this reason, we focus on empirical studies examining the TCO of clean vehicles published in the past five/six years.

The economic analyses of transportation services analyzed by existing studies are differentiated by the type and size of vehicles. These include studies analyzing the economics of passenger cars (Lévay et al., 2017; Breetz & Salon, 2018; Palmer et al., 2018; Weiss et al., 2019; Ouyang et al., 2021; Ajanovic & Haas, 2021; Hamza et al., 2021), commercial vehicles (Falcão et al., 2017), buses (Li & Kimura, 2021; Comello et al., 2021; Ally & Pryor, 2016), trucks (Vora et al., 2017) and combination of different vehicles (Gambhir et al., 2015). In general, most of these studies show that the TCO of an average passenger EV is currently higher than an equivalent ICEV. Table 3 below presents several studies that show the cost disadvantage of alternative (or cleaner) vehicles as compared to their conventional counterparts.

However, the EVs are projected to be cost-competitive in the future because of series of efforts and commitments in place to improve EV's battery technology, which constitute a major cost of EVs. Meeting the climate change targets, mainly the net-zero carbon target by the mid-

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[10] In the case of transportation service providing firms, the TFP can be measured dividing output (e.g., transport sector value added) by a weighted set of inputs (e.g., labor hours, fuel, equipment, and materials). It can also be measured as average revenue per passenger-mile for passenger transportation, and average freight revenue per ton-mile for freight transportation (Davis & Boundy, 2021; BTS, 2021).

[11] Road transportation vehicles can be broadly categorized as conventional internal combustion engine vehicles (ICEVs) and alternative electric vehicles (EVs). The ICEVs are powered by fossil fuels (gasoline, diesel, LPG, and natural gas) and in some cases, biofuels (biodiesel and bioethanol) are blended with fossil fuels. The EVs are powered by electricity. There are two basic types of EVs: all electric vehicles and plug-in hybrid electric vehicles (PHEVs). All electric vehicles include Battery Electric Vehicles (BEVs) and Fuel Cell Electric Vehicles (FCEVs). The PHEVs run on electricity for shorter ranges, then switch over to an ICE running on gasoline when the battery is depleted. PHEVs could also use hydrogen in a fuel cell, biofuels, or other alternative fuels as a back-up instead of gasoline.

century, require a massive investment in the transportation sector. The sector needs to undergo a transition towards massive electrification across the different modes; production of zero-carbon synthetic fuels derived from renewable energy and massive improvements in vehicle EE. Some studies project when alternative or cleaner (i.e., EVs, BEVs, hydrogen vehicles) vehicles will be economically competitive through technological innovations and market development. Whiston et al. (2021) find that FCEVs would be competitive with conventional vehicles by 2035. Ouyang et al., (2021) also argue that small BEVs will reach parity with ICEVs by 2025, and medium-sized and large BEVs will do so around 2030. Even though BEV and PHEV purchase costs will fall by 31%–36% and 16%–18%, respectively, between 2020 and 2030, most EV models will still not reach purchase cost parity by 2030.

A few studies report cost-effectiveness of alternative or cleaner vehicles, particularly when they account for environmental benefits. Holland et al. (2021) find that electric buses are cheaper than diesel and CNG buses in urban metro areas in the US. For example, compared to new diesel or CNG buses, the per-mile net benefit (or cost savings) of an electric bus is \$0.04 and \$0.01, respectively, on average. Although significant spatial heterogeneity exists across different urban areas within the country, the environmental benefit of electrifying the entire bus fleet in Los Angeles relative to a new diesel fleet is estimated at US\$65 million per year; relative to a diesel bus, the benefits of purchasing and operating an electric bus over its lifetime is about US\$8,000 at a 3% discount rate. Lebeau et al. (2013) find that electric buses are more cost-effective than the existing diesel buses in Belgium, mainly because the TCO is around €730 thousand for electric buses and €803 thousand for diesel buses. In Malaysia, Teoh et al. (2018) find that the annual cost (i.e., energy, maintenance, and fuel costs) for electric buses is around US\$4.5 million, which is 68% less than that of a diesel bus. Majumder et al. (2021) report similar results for India. Depending upon how electricity is supplied (grid or solar off-grid or combination of both), they find that the present value of the total cost of an electric bus system by the end of 2030 ranges from US\$314 thousand to US\$370 thousand. In contrast, it is US\$377 thousand for the conventional diesel bus system. Borén (2020) finds that electric and alternative fuels (biogas and HOV) fueled buses in Sweden will be cheaper than diesel vehicles if the societal cost of emissions and noise are considered.

Although some existing studies above, presented in Table 3, show that alternative or green/clean transportation systems are relatively cheaper than existing fossil fuel-based transportation services, the actual deployment of clean vehicles faces several barriers. First, the economic analysis of the green/clean transportation system may not necessarily reflect the true economic costs as these analyses include financial incentives (i.e., subsidies provided under the various government programs to promote green/clean transportation systems). This issue is highlighted in some of the studies presented in Table 3 (e.g., Li et al., 2021b). Some studies include environmental benefits in the analysis (e.g., Holland et al., 2021). Including the environmental costs of fossil fuels or the environmental benefits of clean fuels is a correct approach in an economic analysis of clean/green transportation. However, private vehicle owners do not account for these social benefits while making a vehicle purchase decision. Some of these studies (e.g., Ruffini & Wei, 2018; Ouyang et al., 2021) make optimistic assumptions on the technological evolution of clean/green transportation systems, which could be unrealistic due to series of uncertainties around the assumptions. Moreover, even if some cleaner vehicles are economically attractive under a certain set of data and assumptions, adoption of these vehicles may not occur due to several barriers, including lack of infrastructure (e.g., charging stations to electric vehicles).

Table 3. Selected empirical studies assessing the economics of alternative vehicles

Study	Methods	Findings and argument
Holland et al. (2021)	Damage valuation using the IAM model, and the social cost of carbon is used for an electric urban bus in the USA.	Compared to a diesel bus fleet, the environmental benefit of operating an electric bus fleet is about US$65 million per year in Los Angeles and above US$10 million per year in six other metropolitan areas in the USA. Relative to diesel, the NPV benefit of an electric bus is positive in about two-thirds of urban counties. Relative to CNG, the NPV benefit is negative in all counties.
Whiston et al. (2021)	Based on 31 experts' assessments of expected future costs and capacities of storage systems in the US.	Given technical and fuel price uncertainty, FCEV costs ranged from US$0.38 to 0.45 per mile in 2020, US$0.30 to 0.33 per mile in 2035-2050, and US$0.27 to 0.31 per mile in 2050. Depending on fuel, electricity, and battery prices, FCEVs could compete with conventional and alternative fuel vehicles by 2035.
(Li et al. (2021b)	A life cycle cost model is used for BEV and FCEV in China.	BEVs and FCEVs in most cities in China are incomparable with conventional ICEs in terms of tangible cost. But government subsidies, purchase and driving restrictions, and environmental taxes, greatly increase the intangible and external costs of CVs, making consumers more inclined to purchase BEVs and FCEVs.
He et al. (2021)	A levelized cost of driving for light duty FCEV is applied over the period of 2017 to 2030 in China	Assuming decreased costs for both hydrogen production and FCVs in 2030, the levelized cost of driving for wind-electrolysis light-duty FCEVs pathway (US$0.31/km) could approach that for gasoline (US$0.29/km) and BEVs (US$0.30/km). The vehicle purchase cost ranges from US$23,999 for ICEV to 55,292 for BEV in the reference

		case for 2017 model, while purchase cost ranges from US$25,612 for ICEV to 37,039 for BEV for 2030 model.
Comello et al. (2021)	The time-driven life-cycle cost model is used for mobility services of an electric transit bus in the USA.	Based on the LCC of diesel and battery-electric transit buses, electric buses entail higher upfront acquisition costs, but they obtain lower LCCs once utilization rates exceed only 20% of the annual hours (i.e., 1300 hours).
Ouyang et al. (2021)	The consumer oriented TCO approach is used for ICEVs, PHEVs, and BEVs in China from 2020 to 2030.	Under the 5-year holding period scenario, small BEVs will reach parity with ICEVs in terms of the TCO by 2025, while medium-sized and large BEVs will do so around 2030. Even though BEV and PHEV purchase costs will fall by 31%–36% and 16%–18%, respectively, between 2020 and 2030, most EV models will still not reach purchase cost parity by 2030.
Tarei et al. (2021)	Best-Worst Method is used to filter out the critical EV barriers out of a total of eighteen barriers obtained from the existing literature in India.	Five main EV barriers are technical, infrastructure, financial, behavioral, and external barriers. Technical barriers include relatively lesser known as a technology than ICEV, unknown driving and durability range, unreliability of suppliers, and not well-known development of alternative fuel technology. Infrastructure barriers include shortage of charging stations, low availability on maintenance, service, and repair, lack of EV manufacturers, and unavailability of reliable electricity. Financial barriers include high upfront purchase price, and unknown resale value and TOC. Behavioral barriers include consumer perception on EV such as lack of awareness, skepticism on safety and reliability, and perceived benefits, and dealer understanding reluctance to push EVs. External barriers include dependence on external sources for raw materials, wastage and recycling of battery, and limited EV incentives and advertisement by government.
Ajanovic & Haas (2021)	TCO approach is used for FCEV.	Based on a driving range of 12,000 km per year over 7 years, the TCO of FCEVs is currently very high, about three times, compared to conventional ICEVs. Three major challenges that the passenger FCEVs to be in strong competition with conventional ICEVs and BEVs are reduction of investment costs of cars, infrastructure development, and stable policy framework conditions.
Hamza et al. (2021)	Bottom-up approach for TCO is applied in the USA.	In 2018, the costs of plug-in vehicles (BEVs and PHEVs) are approximately US$7,000, 8,000 and 11,200 more than the conventional ICEVs in the size categories of car, crossover and SUV, respectively. Even with expected reduction in battery cost in 2030, the plug-in vehicles are approximately US$1,800, 2,500 and 3,500 more than conventional ICEVs for car, crossover and SUV, respectively.
Hasan et al. (2021)	TCO per km approach is used in New Zealand.	The TCO of new EV is much higher compared to conventional ICEV mainly due to high initial purchase price and the cost of battery replacement. However, with proposed "clean car discount", the TCO per km of EV vehicle (Nissan Leaf) is expected to reduce by 6 NZ cents and becomes competitive with new conventional ICEV (Toyota Corolla). In 2018, the TCO per km of new EV is 45.6 NZ and new conventional ICEV is 38.8 NZ cents.

Li & Kimura (2021)	TCO per km approach in EVs in ASEAN countries.	Based on TCO per km, the FCEVs are the least competitive option among the alternative powertrains. BEVs appear to be the most competitive, except for passenger vehicles. The TCO per km varies from low US$0.55 in Thailand to high US$0.75 in Malaysia for FCEV passenger cars.
Mustapa et al. (2020)	TCO analysis is used in Malaysia.	The TCO per km of Nissan leaf (BEV), BMW i3s (BEV), Hyundai Ioniq HEV plus (PHEV), Honda jazz 1.5 (PHEV), and Perodua Myvi 1.5 High AT (ICEV) is 1.75, 2.5, 1.0, 0.71, and 0.52, respectively, based on 28,188 km over 20 years period. The TCO per km of the BEVs are higher compared with the PHEVs and the ICEV. This implies that the cost of owing EVs in Malaysia is not competitive compared to the HEVs and the ICV.
Cox et al. (2020)	TCO analysis is used in Europe.	Vehicles with smaller batteries and longer lifetime distances have the best cost and climate performance. If a very large driving range is required or clean electricity is not available, hybrid powertrain and compressed natural gas vehicles are good options in terms of both costs and climate change impacts. Alternative powertrains containing large batteries or fuel cells are the most sensitive to changes in the future electricity system as their life cycles are more electricity intensive.
Nassif & Almeida (2020)	Advanced Vehicle Simulator (ADVISOR) is used.	The costs and fuel economy results of the different configurations (different degree of hybridization, DOH) of FCEV are compared to those of the original vehicle. The configuration with the highest DOH (61.2%) show an 8.3% increase in fuel economy and a total cost reduction of 13.2% compared to the original vehicle. In 2019, the purchase price of Nissan Leaf (40 kWh) and Hyundai Ioniq Electric ranges from US$30,000 to 37,000, and Toyota Mirai FCEV costs US$58,500.
Dreier et al. (2019)	Advanced Vehicle Simulator (ADVISOR) is used in Brazil.	Analyzing the influence of passenger load, driving cycle, fuel price, and four different types of buses on the cost of transport service for one bus rapid transit route in Curitiba, Brazil, the lowest fuel cost ranges for the passenger load are plug-in (0.198–0.289 US$/km), followed by two axle (0.255–0.315 US$/km), articulated (0.298–0.375US$/km), and conventional (0.552–0.809 US$/km) buses. The capital cost of bus ranges from US$103,669 to 276,563.
Bekel & Pauliuk (2019)	Life cycle assessment is used in Germany.	Considering mid-size vehicles like the VW e-Golf, BEVs have today a better environmental and financial performance than FCEVs. However, the range of the BEV is lower than the range of the FCEV (200km vs 530 km) and both technologies have different stages of maturity.
van Velzen et al. (2019)	Literature review and interview methods is used.	The production costs of EVs will go down in the future, but this does not mean that the TCOs of EVs will continue to go down as well. It is argued that EVs are sold at or below production costs. This is a common strategy in the automotive industry, as initial investments are high, and no mark-up is covered by the BEV's retail price. This may change in the future and BEV retail price may increase to recoup investments. This directly influences the TCO of BEVs in a negative way, suggesting that the ongoing TCO of BEVs in the future may not decrease.

Thompson et al. (2018)	Design for Manufacture and Assembly (DFMA®) analysis is used.	Total system cost ranges from US$50/kWnet to produce 100,000 FCEVs to US$45/kWnet to produce 500,000 FCEVs per year in 2017 based on the comparison with the fuel cell system in the commercially available Toyota Mirai. Decreasing system cost to US$30/kWnet will be needed for cost competitiveness with ICEVs.
Ruffini & Wei (2018)	Learning The learning rate approach and life cycle cost analysis are used.	The fuel cell system is the key factor in making FCEV life cycle costs comparable to ICEV costs. With an 18% learning rate, FCEVs are estimated to be cost-competitive with ICEVs by 2025, but with an 8% learning rate, this cost-competitive point is pushed out almost 25 years.
Breetz & Salon (2018)	TCO approach is used for conventional, hybrid, and electric vehicles in 14 cities in the USA from 2011 to 2015.	BEV cost substantially more than the conventional and hybrid vehicles mainly due to their higher purchase price and rapid depreciation outweighed its fuel savings. Based on 10 years TCO, to bring the EV (Nissan Leaf) to cost parity with the ICEV (Toyota Corolla), the Leaf's purchase price would need to drop by US$5,091 (15%) under full electricity prices, US$3,192 (10%) with half-priced electricity, and US$1,292 (4%) with free electricity.
Ally & Pryor (2016)	TCO based on life cycle assessment of FCEV (buses) compared with diesel, CNG, and hybrid buses in Australia from 2012 to 2014.	At the current level of capital and operation costs and with the current performances of hydrogen and fuel cell technologies, the TCO of a fuel cell bus is 2.6 times that of a conventional diesel bus.
Simmons et al. (2015)	Vehicle model-specific approach and sales-weighted average approach is used in the USA.	Consumers realize, on average, 17.3% fuel economy improvement and save US$1,070. Most new technologies (hybrid electric vehicles) become financially attractive to consumers when average fuel prices exceed US$ 5.60/gallon, or when annual miles traveled exceed 16,400.
Gambhir et al. (2015)	MACC and decomposition analysis is used for low-carbon vehicles in China.	The substitution of low-carbon fuel-driven trucks (fuel cell and battery) for their business-as-usual alternatives (ICEV) results in cost savings by 2050. In contrast, the low-carbon fuel-driven passenger cars are, in most cases, significantly costly -- 20 to 60 percent more expensive than ICE vehicles in 2050.

4.3 Power Sector

The electric power sector offers the highest opportunities to invest in green or clean infrastructures, particularly renewable electricity (RE) generation technologies, including hydro, solar, wind. The critical question is whether the investments in green/clean power generation technologies help reduce the total electricity supply costs. The answer is mixed (please see Table 4). In some cases, the expansion of renewable electricity reduces the cost of electricity supply and eventually electricity prices. On the other hand, it would result in the opposite in many cases. In the following section, we present existing literature analyzing the impacts of renewable or clean electricity generation technologies on electricity supply costs and prices.

Even though costs for renewable sources for electricity generation are falling rapidly, their initial investment (capital) costs are still high as compared to the conventional technologies to produce electricity (e.g., natural gas combined cycle technology). On a levelized cost basis, renewables are becoming more and more competitive with conventional power generation technologies (Timilsina, 2020). However, intermittent renewables cannot be compared with existing technologies that supply uninterruptible electricity (or firm capacity). Therefore, increased penetration of renewable energy technologies is likely to increase the average costs of electricity supply unless the cost of energy storage drops significantly. Even if carbon pricing is introduced to penalize fossil fuels, renewables could still increase the average costs of electricity supply of a grid (Steckel & Jakob, 2018).

There are two types of studies analyzing the relationships between the penetration of renewable electricity into a grid and its average costs of electricity supply. The first set of studies conduct ex-ante analysis using power sector expansion models, which use optimization techniques. These models find the least-cost mix of electricity generation resources for a given time-horizon satisfying all constraints specified by the modelers. For example, studies such as Mai et al. (2021), Liang et al. (2019), and Kumar (2016) have employed this approach to examine the effects of expanding renewable electricity on the costs of electricity supply from grids. The second type of studies are ex-post studies using econometric techniques to relate the expansion of renewables with electricity prices. Würzburg et al. (2013) and several studies referred by them fall into this category.

Many studies have investigated the impacts of increased penetration of renewable electricity on the average costs of electricity grids and, ultimately, electricity prices. In Italy, Vellini et al. (2020) find that replacing coal-fired power plants with renewables would increase the electricity supply costs because the benefits from coal phase-out are smaller than electricity system costs due to increased intermittent renewables and natural gas. Studies, such as Adelman & Spence (2018), argue that renewables pose a threat to the viability of base-load generation in the long term; therefore, cheaper, and relatively cleaner fossil-fuel sources, such as natural gas, are needed for the reliable supply of electricity for many decades to come. Studies, such as Liang et al. (2019) and Kumar (2016), also find that large-scale integration of renewable energy would cause an increase in electricity system costs and the price. However, if the declining trends of costs of renewables and storage technologies continue, and if the value of carbon reduction is accounted for, integration of renewable technologies may not increase the long-term costs of electricity supply from a grid.

Mai et al. (2021) examines the impacts of meeting renewable energy portfolio standards introduced or planned by various states in the USA, using National Renewable Energy Laboratory's electricity capacity expansion model. They find an expansion of non-fossil electricity generation that accounts for 45% of the total generation would increase electricity system costs by 0.4% to 0.8% depending upon the discount rate. If the share of non-fossil fuels in the total generation reaches 78%, the system costs will increase by 2.7% to 3.6%. Using an energy system model, LEAP, Liang et al. (2019) find in China that increasing the share of renewable electricity capacity by 23.6 percentage point in 2050 (from 27.1% in BAU scenario to 50.7% in RE expansion scenario) would increase the electricity system costs by 7%. Using the LEAP model, Kumar (2016) finds that expanding renewables in Indonesia to share 40% of the total electricity generation by 2050 (or increasing renewable capacity by 1.3 times from the business-as-usual case) would increase the electricity system costs by 40% from the BAU case. The same study finds that increasing renewable capacity to account for 39% of the total capacity by 2050 (increasing renewable capacity by 3.4 times from the BAU case) would increase the electricity system costs by 82% (from the BAU case) in Thailand.

Schulte & Fletcher (2021) argue that the cost of attaining RE penetration in a grid above a threshold or inflection point (the point at which the penetration of RE begins to exceed hourly load) gets very expensive for the retail utility customers. At this point, REs must be turned down or curtailed. This situation results in diminishing returns for REs. This could be a technical barrier to achieve a 100% clean energy goal for an electricity grid.

In certain circumstances, the expansion of renewable electricity could reduce the average costs of an electricity supply system or grid and electricity prices. Such cases arise when the electricity market is well-interconnected and is fully deregulated; electricity prices are based on short-run prices, which reflect variable costs of the network at a given time, such as day-ahead price. In the EU electricity market, which represents a well-interconnected system of electricity grids, increased penetration of renewables could reduce electricity prices at certain points in time (i.e., certain hours in a day). Würzburg et al. (2013) list several studies that report a reduction of short-term costs of electricity in the EU and the corresponding prices due to the increased penetration of renewables. However, some caution needs to be taken as to whether RE sources have enjoyed subsidies or premium prices (feed-in-tariff). The price reduction could be a result of these subsidies. On the other hand, one could also argue that if carbon prices are introduced, the price reduction effects of RE sources hold even if they do not receive any subsidies. It should also be noted that the studies cited in Würzburg et al. (2013) are relatively old (about a decade at least); costs of electricity from RE sources were much higher a decade ago than now. The average prices of electricity are higher now than a decade ago.

Table 4. Selected empirical studies showing impacts of renewable energy on electricity supply costs

	Methods	Findings
Schulte & Fletcher (2021)	Useful clean energy analysis in the USA.	Due to intermittency, RE leads to diminishing returns for useful clean energy. Every utility has a hypothetical "inflection point" where increasing high levels of renewables begin to exceed hourly load and must be turned down or otherwise curtailed. Utilities would "stall out" in useful energy when RE reaches levels at about 60 %–80 % of their annual load.
Mai et al. (2021)	Optimization-based capacity expansion model and Regional Energy Deployment System are applied in the USA from 2020 to 2050.	The expansion of non-fossil electricity generation to account for 45% of the total generation would increase electricity system costs by 0.4% to 0.8% depending upon the discount rate. If the share of non-fossil fuels in the total generation reaches 78%, the system costs would increase by 2.7% to 3.6%.
Vellini et al. (2020)	Cost-effective estimation of mitigation costs of the power sector in Italy.	The mitigation costs CO_2 emissions reduction of the Italian electricity sector in 2030 vary widely ranging from null up to €140 per ton. The larger the use of gas and electricity generation from the solar source, the higher are the mitigation costs. The complete phase-out of coal exhibits the highest migration costs that are not as beneficial from an economic perspective.
Liang et al. (2019)	A simulation method using the LEAP model is applied in China from 2015 to 2050.	Increasing the share of renewables in total electricity generation capacity from 27.1% to 50.7% in 2050 would increase the electricity system costs for the 2015-2050 period by 7%. It would reduce 41% of the power sector's CO_2 emissions in 2050.
Adelman & Spence (2018)	Low-cost electricity generation optimization is applied in the USA from 2016 to 2031.	The low natural gas prices—below US$3.50/MMBtu through 2023 and below US$4.50/MMBtu through 2031—inhibit the construction of new renewable capacity. The significant impact of inexpensive gas is likely to be limiting the entry of renewables. This is mainly due to two phenomena: 1) renewables pose a much greater threat to the viability of base-load generation in the longer-term than a natural gas-fired generation; and (2) when gas prices set market prices, they also determine the economics of renewables, and thereby the volume of new renewable capacity that enters the market.
Kumar (2016)	A simulation method using the LEAP model is applied in Indonesia and	In implementing renewables at a large scale in both these countries the cost of production increases substantially.

	Thailand from 2020 to 2050.	For example, the total electricity production costs in 2050 increases by 40% in Indonesia and by 82% in Thailand.
Würzburg et al. (2013)	A multivariate regression model is used in Germany and Austria.	The changed electricity mix due to the German nuclear electricity generation exit did not affect the size of the merit order effect, where price decreases occur because (additional) RE based electricity bids into the market at lower marginal costs. It also lists several earlier studies analyzing the impacts of RE in the EU electricity market.

4.4 Buildings sector

Several studies examine the economics of energy-efficient technologies, particularly heating and cooling devices, insulations, lighting devices, cooking appliances, electronic and computer devices in residential and commercial buildings. These technologies reduce the demand for energy and save energy costs; they reduce emissions of GHGs and local air pollutions (e.g., indoor air pollution). Globally, a large investment is made annually in clean technologies in buildings. In 2018, it accounted for the largest share (58%) of the total US$139 billion invested in energy-efficient technologies globally[12] (IEA, 2019). In this section, we highlight recent literature on the economics of energy-efficient technologies in the buildings sector. Several empirical studies show significant savings in energy consumption and energy bills in the household sector (a sub-set of the buildings sector). Reviewing 100 existing studies analyzing EE improvements in the household sector, McAndrew et al. (2021) report a wide variation of reduction in electricity use ranging from a low of 0.5% to a high of 80% of total electricity use, with a median value of 7.9%. Some selected studies are summarized in Table 5.

With an econometric analysis using data from a field survey of residential energy consumption in Amman and Zarqa regions of Jordan, Abd Alfattah et al. (2017) find the net saving by implementing EE measures (refrigerators and freezers) from 2011 to 2020 ranges from 4,451 GWh to 17,807 GWh depending upon different scenarios, which is equivalent to saving of electricity bills by Jordanian Dinar 320 million to 1,282 million, respectively, roughly US$450 million to 1,808 million in the current exchange rate. Using a multivariate statistical approach on data collected from 1,400 dwellings in France, Belaid et al. (2021) find that low-temperature and condensing boilers, as well as floor insulation, are the most cost-effective EE measures. The

[12] EE investments in buildings sector include building envelopes, heating, ventilation and air conditioning (HVAC), appliances, and lighting.

percentage of total energy saved by using each type of these measures is about 29% for floor insulation and 38% for both low-temperature and condensing boilers. The cost-effectiveness of energy renovation measures is widely dependent on energy prices. Through the exchange program of replacing inefficient bulbs with LED bulbs of 181 households in low-income areas of Salt Lake City, Utah, Witt et al. (2019) estimate that residents saved approximately US$18,219 annually in electricity bills, an equivalent of about US$100 per household. This translates to savings of about 2.6% of the total electricity consumed by each household. Using artificial neural networks and empirical models on a sample of 16 household refrigerator replacements under the Appliance Replacement Offer in low-income households in Australia, Ren et al. (2021) find most low-income households save, on average, 53% of their energy consumption in refrigeration through the replacement of old refrigerators with energy-efficient ones. However, they also find that some portion of the energy savings could be compromised if the size of a new refrigerator is larger than the old one.

It is not necessarily true that the adoption of clean or energy-efficient technologies reduces energy consumption. Employing a Logit regression model on 2010 household budget survey of Ireland, Chesser et al. (2019) show households that have adopted micro renewable energy systems, such as PV panels, micro wind turbines, solar thermal water heaters, wood pellet boilers, geothermal heat pumps and combined heat and power (CHP) units, have increased their electricity use. Using micro-level data from the National Survey of Family Income and Expenditure in Japan, Inoue and Matsumoto (2019) find households began spending more electricity on space cooling and food preservation after the implementation of the EE program that sets efficiency standards for major home appliances. They find that a large size and increased stock of home appliances contributed to an increase in the electricity use. Based on a controlled field experiment of 154 apartments in the Greater Boston area in 2011, Tiefenbeck et al. (2013) find residents who received weekly feedback on their water consumption (a proxy for environmental measures) lowered their water use by an average 6%, an equivalent of 0.5 kWh/person/day, but at the same time, they increased their electricity consumption by 5.6%, an equivalent of 0.89 kWh/person/day. They estimated that the energy saved by reduced (hot) water consumption is offset by the increased electricity consumption by nearly a factor of two. Jacobsen et al. (2012) also find that households in Memphis, Tennessee using the minimum threshold level (150 kWh per month) of green electricity program (a proxy for environmental measures) have increased their electricity consumption by 2.5%.

Several studies use economic or financial analysis to evaluate investments in clean technologies (EE and solar home systems). Below we briefly highlight the findings of the studies by technology types. First, we discuss energy utilization technologies followed by solar home systems.

Analyzing data from a national survey in Ghana, Opoku et al. (2019) estimate that the cost savings from reduced electricity consumption by adopting energy-efficient air conditioning (AC) systems is about US$1.96 billion from 2018 to 2030. Suástegui Macías et al. (2018) report that 39% of households in Mexicali, Mexico, use oversized ACs and that replacing these ACs with more energy efficient ACs would save 32% in energy consumption annually.

By setting optimal HAVC (heating, ventilation and air conditioning) temperature setpoints in office buildings in seven climate zones across the USA, Papadopoulos et al. (2019) show that up to 60% of annual HVAC-related energy is saved without compromising occupants' thermal comfort. Wu et al. (2018) find 13.1% to 14.7% of energy savings with more efficient HVAC (e.g., ground-source heat pump with two and three boreholes) compared to conventional HVAC (e.g., air-source heat pump) in a single-family house in Maryland, USA. The estimated value of operation cost saved is about US$186 to US$191 per household annually. However, for the same net electricity generation, the ground-source heat pumps required US$7,934 (2 boreholes) and US$9,739 (3 boreholes) higher initial cost than the air-source heat pump.

Some studies examine net energy savings (or net benefits) due to the adoption of clean energy supply systems in buildings. Timilsina (2021) finds that the benefits of distributed photovoltaics (DPV) system in Bangladesh would be more than 1.5 times in residential and commercial buildings. A study for the Waukegan area of Illinois, Baek et al. (2020) find that electricity bill savings from solar PV and energy storage system pay back the investments in 11–12 years, as most homes and businesses in the region could cut their electricity bills by more than half. Park et al. (2019) show that the refrigerator electricity consumption can be reduced by about 50% and 70% using commercially available energy-efficient components at an incremental cost of about US$45–US$60 and US$100–US$120 per unit, respectively. They also find that the total annualized cost of an off-grid solar home system together with refrigerator can be decreased by about 50%, if the most efficient refrigerator is used, mainly due to the additional cost of the efficient refrigerator is significantly lower than the cost savings due to smaller capacity requirements for panels and batteries. In Australia, Roberts et al. (2019) find 25% to 43% energy savings by individual households when solar PV and battery systems are used for refrigerators. Atikol et al. (2013) find the benefits to cost ratio of replacing storage type electric water heater with solar PV-based water heaters ranging from 3.7 to 7.5 in Cyprus. Retrofitting with EE measures and integrated rooftop PV in the office building in Italy indicated a potential energy savings of 54%, equivalent of 8.8 TWh, with total implementation costs of €19 billion (Luddeni et al., 2018).

Not all case studies or economic/financial analyses show investments in clean technologies reduce the costs of energy services delivered. Whether investments in clean technologies pay off or not depends on several factors. While the detailed discussion of these factors is beyond the scope of this study, we present a few examples below. Energy price is the key factor influencing the economics of clean technology investment in the buildings sector. Lohwanitchai and Jareemit (2021) find that the investment in energy-efficient technologies, such as high-performance glazed windows in small office buildings in Thailand, is unprofitable under current economic conditions. The electricity savings from building envelopes (e.g., external insulating wall) of a typical apartment in the hot and humid climate of Israel is only one-third of the total retrofit cost (Friedman et al., 2014). By retrofitting the HVAC system and insulation of office buildings in China, annual electricity consumption savings of 75% compared to the national average can be achieved (Zhou et al., 2016). They find that electricity consumption savings of 53% for heating and 64% for cooling with similar scale buildings elsewhere in the country. They also find that 30% of the operation cost can be saved through replacing HVAC system operation with water storage system (energy-efficient retrofitting). While retrofitting wall insulation and replacing old ACs with energy-efficient ones offer higher financial viability, replacing windows is not in the case of residential villas in Dubai (Rakhshan and Friess, 2016). Likewise, retrofitting heating system is cost-effective, while building envelops (e.g., retrofitting external walls) is not economically beneficial in residential buildings in Beijing, China (Liu et al., 2018).

Table 5: Selected studies on the economics of clean technologies in the buildings sector

Study	Method	Main findings and argument

Belaid et al. (2021)	A multivariate statistical approach and cost benefit analysis are used to 1,400 dwellings of the French residential sector in 2013.	Low-temperature and condensing boilers, as well as floor insulation, are the most cost-effective EE measures. Also, the cost-effectiveness of energy renovation measures is widely dependent on energy price and the discount rate.
Lohwanitchai and Jareemit (2021)	A cost-benefit analysis is used for three representative six-story office buildings in Thailand.	The investment of high-performance glazed-windows in the small office buildings is unprofitable (NPVs = −14.77–−46.01).
McAndrew et al. (2021)	A literature review based on 153 papers published between 1990 and 2019 is used to examine the EE interventions in advanced economies' households.	Review of one hundred papers reported a positive impact of EE intervention, i.e., a reduction in electricity use by households, ranging from 0.5% to 80%, with a median measured electricity reduction of 7.9%.
Ren et al. (2021)	The artificial neural networks and empirical models are used for a sample of 16 household refrigerator replacements under the Appliance Replacement Offer in low-income households in Australia.	Most households achieved substantial energy savings from the replacement program. The average annual energy cost savings of refrigerator replacements is 53%. However, the energy savings could be compromised if the size of the new refrigerator were much larger than the old unit.
Timilsina (2021)	Economic analysis of distributed PV (DPV) in the residential, commercial, and industrial sectors using a model that allows electricity exchange between the consumers and the national electricity utility	The benefits of DPV to the consumers are more than 1.5 times as high as their costs
Baek et al. (2020)	An hourly load profile simulation is used for solar-plus-storage applications for three types of buildings: secondary school, supermarket, and a typical single-family home in Waukegan, Illinois.	In most cases, investing in solar PV resulted in significant electricity bill savings for consumers that paid back the initial capital cost of the system in 11–12 years. Most homes and businesses in the Waukegan area in Illinois could cut their electricity bills by more than half.
Chesser et al. (2019)	A Logit regression model is used based on 2010 household budget survey of Ireland.	Households that have adopted the micro- renewable energy systems do not result in a reduction of electricity use, rather it increases electricity use.
Inoue and Matsumoto (2019)	A conditional demand analysis is used based on micro-level data from the National Survey of Family Income and Expenditure in Japan.	Although EE of home appliances significantly improved after the implementation EE program (Top Runner), households began spending more electricity on space cooling and food preservation.
Opoku et al. (2019)	Technical data from a national survey is analyzed to study EE of ACs used in the offices of public and commercial buildings in Ghana.	The estimated annual electricity savings potential is 260 GWh in 2020 and 1,770 GWh in 2030; and savings of US$1.96 billion during 2018-2030 using more efficient inverter ACs instead of ACs currently in use.
Roberts et al. (2019)	Financial analysis is used for PV solar and battery storage system in 5 Australian apartments.	There is a financial benefit to deployment of embedded networks with combined solar and battery storage systems for apartment buildings. The cost of these measures ranges from 400 – 750 AUD per kWh for embedded networks as compared to 750–1000 AUD per kWh for individual household system.
Papadopoulos et al. (2019)	Simulation-based multi-objective optimization is used to fine-tune HVAC setpoints of office buildings in 7 climate zones across the US.	Locations with mild climates, such as San Francisco, CA, can realize up to 60% of annual HVAC-related energy savings without compromising the occupants' thermal comfort.

Park et al. (2019)	A bottom-up approach is used to estimate the energy consumption and cost of small (50 L and 100 L) refrigerators.	The refrigerator electricity use can be reduced by about 50% and 70% using commercially available energy-efficient components at an incremental cost of about $45–$60 and $100–$120 per unit, respectively.
Witt et al. (2019)	Based on a project of exchanging LED bulbs for 181 households in low-income areas in Salt Lake City, Utah over the period of 8 months.	A saving of approximately US$18,219 is estimated in electricity bills for the residents. This is an equivalent of about US$100 savings per household per year.
Alfattah et al. (2017)	An econometric approach is used for field survey of residential sector in Amman and Zarqa, Jordan.	The net saving by implementing EE measures (refrigerators) from 2011 to 2020 is estimated at about 4451 GWh which translate into saving of electricity bills by 320 million JD.
Krati and Dubey (2017)	An optimization analysis is used to evaluate economic and environmental benefits of a wide range of EE technologies for new and existing buildings in Oman.	The implementation of a government funded large scale energy retrofit program for the existing residential building stock is highly cost-effective and provide a reduction of 957 GWh in annual electricity consumption.
Jacobsen et al. (2012)	A cross-sectional probit model is used based on a green electricity program for households in Memphis, Tennessee from 2003 to 2008.	Participating households that enroll at the minimum level (i.e., 150 kWh/month) of green electricity program, increase their electricity consumption by 2.5%.
Kneifel (2010)	A life cycle cost analysis is used for a total of 576 energy simulations are run for 12 prototypical buildings in 16 cities, with 3 building designs for each building-location combination.	EE technologies, such as installation of smaller and cheaper HVAC equipment, can be used to decrease energy use in new commercial buildings by 20-30% on average and up to over 40% for some building types and locations.

4.5 Agriculture sector

Unlike in other sectors, studies examining the impacts of the adoption of clean energy technologies are limited in the agriculture sector. We are highlighting here a few studies available. Efficiency for Access Coalition (2019) reports that an application of solar water pumps for irrigation in remote areas increases farmers' yields by as much as three-fold compared to relying solely on rainfall. Based on an interview of 375 solar water pump users in three African countries (Kenya, Tanzania, and Uganda), the report also finds that majority of the farmers (75%) saying their productivity, such as working less time, increased yields, lower farm operating costs, and improved income, have increased since using their solar water pump, have increased. In Kenya, solar irrigation shows a high return on investment for horticultural crops compared to diesel alternatives (IFC, 2019). The report finds that for a typical farm in Kenya, the total cost over five years to irrigate one acre area is estimated at US$3,000 when using a solar water pump compared to US$6,000 when using an equivalent diesel pump. Li et al. (2021c) find that the energy use by efficient tillage mode (ridge cultivation with no-till) compared to conventional cultivation with intensive till used in rice fields in Hubei province, China, is 24.7% lower. They find that the net ecosystem economic efficiency, which is calculated by subtracting agricultural inputs cost and carbon cost from grain yield cost, is higher for efficient tillage mode (CHY16,419.9 ± 1,186.0 per hectare) compared to conventional one (CHY9,881.8 ± 217.0 per hectare). A similar study (Nisar et al. 2021) in north-western Indo-Gangetic plains in India shows that efficient no-tillage in maize reduced energy input by 38.4% and 20.1% compared to deep tillage conventional tillage, respectively. They also find that for per tonne of maize grain production, no-tillage saved 1,044 MJ of energy (41%) and increased grain yield by 35.3% compared t conventional tillage. Based on the levelized cost of heat, Çiftçioglu et al. (2020) find that solar thermal (glazed and unglazed solar air collectors) use for drying agriculture, marine, and meat products is cost-effective compared to electricity and fossil fuels (natural gas and LPG) options over the long run. Harchaoui and Chatzimpiros (2017) find a two-fold increase in agricultural productivity between 1961 and 2010, mainly due to energy conversion efficiency improvements and agricultural yields. They also find that the overall livestock energy conversion efficiency[13] increased by 45% from 1961 to 2010; poultry gained 84%, pork 17%, sheep & goat 67% and cattle 27%.

There is a wide difference in agricultural yields between developing and developed countries mainly due to the different levels of technologies used. For example, yields for maize, which is a major staple crop in many parts of sub-Saharan Africa, is about 2 tons per hectare compared to commercial yields of about 8 tons per hectare in the Americas (OECD and FAO, 2018). Also, some clean technologies that are used in milling are not always efficient. For example, PV solar-powered motor system for milling maize in East Africa yields (throughput) about 32.7 kg/hr, which is much lower than the diesel mill, which yields 120 to 150 kg/hr (Efficiency for Access Coalition, 2020).

There are some studies that show the use of digital technologies in enhancing agricultural activities and increasing productivity. For example, David and Grobler (2019) find that households using information and communication technologies (e.g., internet connection and mobile telephone) have positive and significant impacts on their agricultural production (food) in South Africa. Using efficient automated irrigation with advance wireless sensor networks, Nikolidakis

[13] Energy conversion efficiency in livestock production is the ratio of output products to feed inputs both expressed as energy.

et al. (2015) show that the quantity of water used is smaller compared to a conventional one. Based on the data for the 1995-2000 period on 81 countries, Lio and Liu (2006) find a significant positive relationship between the adoption of information and communication technology (ICT) and agricultural productivity. They also find that returns from ICT in agricultural production of the richer countries are about two times higher than those of the poorer countries.

5. Other Factors Driving Voluntary Investment on Green/Clean Technologies

Unlike the general perception that the private sector does not have the interest to invest in new green/clean technologies because of their cost disadvantage as compared to conventional or non-green technologies, some empirical literature shows otherwise. Fama and French (2007) suggest that returns or payoffs are not the only criteria investors use to make decisions on green/clean investments. They argue that investors can buy assets as consumption goods rather than strictly based on returns or payoffs. Many information technology companies (e.g., Google, Facebook, and Twitter) think, during the initial public offerings, that the market for clean technologies will expand in the future even if they are expensive and not profitable for now. Ng and Zheng (2018) provide empirical evidence by comparing the investors' demand between 99 green energy companies and 93 matching samples of non-green energy Fortune 500 firms. The results from their Capital Asset Pricing Model show that green energy portfolios perform comparably or better than a matching non-green energy portfolio. Contrary to the traditional perception that full filling the social responsibility is costly for firms, they find meeting the environmental objectives not affecting firms' market performance.

Using sample data of firms from France, Germany, Italy, the Netherlands, and the UK during the 1985-2011 period, Colombelli et al. (2020) show a positive impact of innovation on the market value of firms. They also find that firms operating in sectors with a high propensity for green technologies yield a significant positive effect along with the stock of green technologies. Based on the financial performance data of 420 energy firms in OECD countries over four years (2013-2016), Hassan (2019) shows that green investment promoted by renewable energy policies in OECD countries improves firms' financial performances.

Palmer and Truong (2017) argue, based on findings of Dangelico and Pujari (2010) and Berrone et al. (2013), green technologies lead to increased profitability for firms.[14] Using 1,020 press releases on green technology products introduced by 79 global firms between 2007 and 2012 and their operational profitability, Palmer and Truong (2017) show a positive relationship between green firm profitability. Palčič and Prester (2020) argue that investments in green technologies or innovations benefit firms as it creates a positive image, which eventually leads to an improvement to their market valuation. Using content analysis of corporate disclosures and financial data of 162 companies listed on the Frankfurt Stock Exchange during the 2007-2016 period, Przychodzen et al. (2018) show firms with green information technology receive higher subsequent returns on assets and the market-to-book values of assets ratios. Saha et al. (2017) suggest that investment in green operations and preservation technology can significantly improve the retailer's financial performance.

Some studies report that green/clean investments do not necessarily improve the financial performance or market values of firms. Investors, such as venture capital firms, would be interested in financing green/clean technologies. But Gaddy et al. (2017) find the other way around. Using the investment data from venture capital firms for the 2006-2011 period, they find clean technologies poorly suited for venture capital investment. They find that materials and hardware industries have a longer payback period than venture capital firms normally expect. The size of capital requirement is normally big for large-scale green/clean technology development or adoption. Venture capital firms see the green/clean technology market as risky and yield low returns because the potential acquirers of these technologies (e.g., utilities and large industrial corporations) are reluctant to buy risky startups. Table 6 provides selected empirical studies on the relationship between green investment and the financial performance of firms or industries.

Table 6. The selected empirical literature on indicating the relationship between clean/green investment and financial performance of firms

	Method	Findings
Chen & Ma (2021)	Regression analysis is applied to 90 Chinese	Green investment has a significant and positive correlation with financial performance. The green investment helps reduce environmental violations and promote

[14] Dangelico and Pujari (2010) show new products based on green/clean technologies result in higher commercial values due to a higher visibility of these products to market actors. Berrone et al. (2013) report new products based on green/clean technologies help attract more customers at higher premiums as a result from a long-term investment on R&D.

	energy firms from 2008 to 2017.	environmental performance, and environmental performance can strengthen the impact of green investment in improving the long-term performance of firms.
Colombelli et al. (2020)	A market value approach is used on a sample of firms from France, Germany, Italy, the Netherlands, and the UK for 1985-2011.	The stringency of the environmental regulatory framework yields a positive and significant impact, as does the stock of green technologies vis-a-vis non-green technologies. Moreover, the environmental regulatory framework positively moderates the positive effect of the stock of GTs. Also, the quality of firms' knowledge stocks is found to positively influence firms' MV.
Hassan (2019)	The natural capital inventory approach is used for 420 energy firms in OECD countries for 2013-2016.	Most renewable energy incentive policies deployed in OECD countries stimulate improved accounting-based measures of financial performance. Substituting renewable energy for fossil fuels, incentivized through RE policies, stimulates improved financial performance of energy companies in OECD countries.
Przychodzen et al. (2018)	Content analysis is used to corporate disclosures and financial data of 162 companies listed on the Frankfurt Stock Exchange for 2007-2016.	Firms with green information technologies are characterized by higher subsequent returns on assets and the market-to-book values of assets ratios. Also, firms introducing green information technology solutions experience permanently lower operating margins and higher costs of goods sold to net sales ratios.
Saha et al. (2017)	An optimization and simulation analysis are used.	Continuous investment in green operations and preservation technology can significantly improve the retailer's financial performance. Results also show that the higher price sensitivity of the market always discourages the retailer from investing in green operations. The retailer needs to invest more in green operations for products with relatively high unit value.
Gaddy et al. (2017)	A comparison is made between the risk and return profiles of clean energy technology investments against those of other sectors in the USA from 2006 to 2011.	The investment in clean energy technologies underperformed compared to investments in other sectors. Among the clean energy technology investments, investments in companies that develop new hardware, materials, chemistries, or manufacturing processes consumed the most capital and yielded the lowest returns.

6. Conclusions and Final Remarks

Investments in green and clean technologies are rapidly increasing, particularly in renewable energy and energy-efficient technologies over the last decade. The primary factors behind the growth of green/clean investment are policies and measures introduced by the government in response to environmental concerns, particularly global climate change. The investment is, however, financed mainly by the public sector. The investment is also financed by the private sector, if they are financially incentivized or mandated through regulatory measures. Will the private sector's engagement continue if there are no financial incentives and regulatory measures? It would if the investment in green/clean technologies reduce production costs and at least do not reduce their profit margins. Is there any empirical or numerical evidence of lowering production costs or improving productivity due to green or clean investments? This paper aims to answer this question through a review of the literature. The review includes ex-post empirical studies as well as ex-ante economic analysis or technological/sectoral modeling.

The findings from the literature appear to be sensitive to two aspects: the type of studies (ex-post vs. ex-ante) and the level of studies (firm or technology). We find in the literature that the ex-post studies to analyze the impacts of clean/green technologies on the production costs or costs of energy service delivery use empirical (econometric) techniques that derive the evidence from observed data. The ex-ante studies use economic/financial (e.g., benefits-cost analysis) at the technology level (e.g., electricity vehicle, refrigerator) or modeling at the network or sectoral level (e.g., solar and technologies for power generation). Most of the ex-post or econometric studies we reviewed, particularly the more recent ones, show a positive relationship between the investments in clean technologies and firms' productivity. Several studies also support Porter's hypothesis that state environmental regulations incentivize innovation, ultimately reducing firms' production costs. Some factors, such as the size of firms and types of investments (i.e., investments for development vs. deployment of technologies), influence the relationships.

At the sectoral level, except for some manufacturing and the buildings sectors, most studies are found to use ex-ante economic analysis or sectoral modeling instead of ex-post econometric approaches. Ex-post studies are focused either on the manufacturing industries or the buildings sector. Ex-ante studies at the sectoral level examine the impacts of green/clean investments on the cost of energy services instead of sectoral productivity. The findings of the studies are mixed. In the buildings sector, existing studies mostly report that adopting clean (both energy efficient and clean energy supply technologies, such as solar home systems) saves energy and energy services. The findings are, however, sensitive to several variables, particularly the cost of technologies and fuels. On the other hand, studies for the transportation sector show that vehicles utilizing cleaner fuels (electricity, hydrogen) are not yet economically attractive compared to conventional vehicles unless their environmental benefits (value of GHG and local air pollution reduction) are quantified. Moreover, even if vehicles using clean or non-fossil fuels are economically attractive under certain assumptions, they face many barriers, mainly supporting infrastructure, such as charging stations for electric vehicles. In the future, however, clean or zero-emitting vehicles could be economically attractive, as shown by several studies. Nevertheless, it depends on the improvement of the battery technology and provision of required infrastructure at much smaller costs than that today. Note that battery constitutes the main cost of electric vehicles.

Studies on the electricity sector suggest that expansion of greener/cleaner renewable energy technologies, which is happening rapidly more recently, also has mixed effects on electricity supply costs from electricity grids. Despite rapid drops in their costs, renewable energy technologies, particularly solar and wind, do not necessarily reduce the average costs of electricity supply because of their intermittency. Moreover, the level of their penetration in most of the countries around the world is still small. If the costs of renewable electricity and storage technologies drop further, increased penetration of renewables will reduce the average costs of electricity.

The literature also reveals that reducing the production costs is not the only criterion for the private sector to decide whether to invest in green/clean technologies. Several studies find that investment in green/clean technologies increases private companies' financial performance as such investments could increase their market value. The private sector could also invest in green/clean technologies to improve their social image because many bigger firms are willing to participate in global efforts to combat climate change. Growing empirical evidence of productivity gains from investments in green/clean technologies and social responsibility, especially in combating climate change, would encourage the private sector to continue or increase investments in green or clean technologies.

The mixed findings of the ex-ante economic analysis and technology/sectoral modeling raise a critical question on whether investments in clean technologies reduce energy consumption and costs of energy service. Answering this question through ex-post analysis using observed data is essential. Otherwise, the investment decisions on clean technologies based on ex-ante analysis do not guarantee to achieve the stated environmental objectives. More ex-post econometric studies using the observed data from the field are essential at the sectoral level, particularly in transportation, power, and buildings sectors.

References

Abd Alfattah, A., Sakhrieh, A., & Al-Ghandoor, A. 2017. Energy efficiency standards and labels for cold appliances in Jordan, *International Journal of Energy Economics and Policy*, 7(3), 95-101.

Adelman, D.E., & Spence, D.B. (2018). U.S. climate policy and the regional economics of electricity generation. *Energy Policy*, 120, 268-275. https://doi.org/10.1016/j.enpol.2018.05.019

Ajanovic, A., & Haas, R. (2021). Prospects and impediments for hydrogen and fuel cell vehicles in the transport sector. *International Journal of Hydrogen Energy*, 46(16), 10049-10058. https://doi.org/10.1016/j.ijhydene.2020.03.122

Aldieri, L., Barra, C., Paolo Vinci, C., & Zotti, R. (2021). The joint impact of different types of innovation on firm's productivity: evidence from Italy. *Economics of Innovation and New Technology*, 30(2), 151-182. https://doi.org/10.1080/10438599.2019.1685211

Aldieri, L., Makkonen, T., & Paolo Vinci, C. (2020). Environmental knowledge spillovers and productivity: A patent analysis for large international firms in the energy, water and land resources fields. *Resources Policy*, *69*, 101877. https://doi.org/10.1016/j.resourpol.2020.101877

Ally, J., & Pryor, T. (2016). Life cycle costing of diesel, natural gas, hybrid and hydrogen fuel cell bus systems: An Australian case study. *Energy Policy*, 94, 285-294. https://doi.org/10.1016/j.enpol.2016.03.039

Arriola-Medellín, A. M., López-Cisneros, L. F., Aragón-Aguilar, A., Romo-Millares, C. A., & Fernández-Montiel, M. F. (2019). Energy efficiency to increase production and quality of products in industrial processes: case study oil and gas processing center. *Energy Efficiency*, *12*(6), 1619-1634. https://doi.org/10.1007/s12053-019-09803-0

Arvanitis, S., Peneder, M., Rammer, C., Stucki, T., & Woerter, M. (2017). Development and utilization of energy-related technologies, economic performance, and the role of policy instruments. *Journal of Cleaner Production*, *159*, 47-61. https://doi.org/10.1016/j.jclepro.2017.04.162

Atikol, U., Abbasoglu, S., & Nowzari, R. (2013). A feasibility integrated approach in the promotion of solar house design. *International Journal of Energy Research*, 37(5), 378-388. https://doi.org/https://doi.org/10.1002/er.3025

Baek, Y., Collingsworth, J., Clemmer, S., Gignac, J., Jacobs, M., Garcia, P., & Sattler, S. (2020). The economic feasibility of solar and storage system in Illinois. *The Electricity Journal*, 33(1), 106689. https://doi.org/https://doi.org/10.1016/j.tej.2019.106689

Bekel, K., & Pauliuk, S. (2019). Prospective cost and environmental impact assessment of battery and fuel cell electric vehicles in Germany. *International Journal of Life Cycle Assessment*, 24(12), 2220-2237. https://doi.org/10.1007/s11367-019-01640-8

Belaïd, F., Ranjbar, Z., & Massié, C. (2021). Exploring the cost-effectiveness of energy efficiency implementation measures in the residential sector. *Energy Policy*, 150, 112122. https://doi.org/https://doi.org/10.1016/j.enpol.2020.112122

Berrone, P., Fosfuri, A., Gelabert, L., & Gomez-Mejia, L. R. (2013). Necessity as the mother of 'green' inventions: Institutional pressures and environmental innovations. *Strategic Management Journal*, 34(8), 891-909. https://doi.org/10.1002/smj.2041

Böhringer, C., Moslener, U., Oberndorfer, U., & Ziegler, A. (2012). Clean and productive? Empirical evidence from the German manufacturing industry. *Research Policy*, *41*(2), 442-451. https://doi.org/10.1016/j.respol.2011.10.004

Borén, S. (2020). Electric buses' sustainability effects, noise, energy use, and costs. *International Journal of Sustainable Transportation*, 14(12), 956-971. https://doi.org/10.1080/15568318.2019.1666324

Boyd, G. A., & Pang, J. X. (2000). Estimating the linkage between energy efficiency and productivity. *Energy Policy*, *28*(5), 289-296. https://doi.org/10.1016/S0301-4215(00)00016-1

Breetz, H. L., & Salon, D. (2018). Do electric vehicles need subsidies? Ownership costs for conventional, hybrid, and electric vehicles in 14 U.S. cities. *Energy Policy*, 120, 238-249. https://doi.org/10.1016/j.enpol.2018.05.038

BTS. (2021). *Transportation Economic Trends*. Bureau of Transportation Statistics (BTS), U.S. Department of Transportation. www.bts.gov/product/transportation-economic-trends

Cao, H., Wang, B., & Li, K. (2021). Regulatory policy and misallocation: A new perspective based on the productivity effect of cleaner production standards in China's energy firms. *Energy Policy*, *152*, 112231. https://doi.org/10.1016/j.enpol.2021.112231

Chen, Y., & Ma, Y. (2021). Does green investment improve energy firm performance? *Energy Policy*, *153*, 112252. https://doi.org/10.1016/j.enpol.2021.112252

Chen, Z., Zhang, X., & Chen, F. (2021). Do carbon emission trading schemes stimulate green innovation in enterprises? Evidence from China. *Technological Forecasting and Social Change*, *168*, 120744. https://doi.org/10.1016/j.techfore.2021.120744

Chesser, M., Hanly, J., Cassells, D., & Apergis, N. (2019). Household energy consumption: A study of micro renewable energy systems in Ireland. *Economic and Social Review*, 50(2), 265-280.

Çiftçioğlu, G. A., Kadırgan, F., Kadırgan, M. A. N., & Kaynak, G. (2020). Smart agriculture through using cost-effective and high-efficiency solar drying. *Heliyon*, 6(2), e03357. https://doi.org/https://doi.org/10.1016/j.heliyon.2020.e03357

Colombelli, A., Ghisetti, C., & Quatraro, F. (2020). Green technologies and firms' market value: a micro-econometric analysis of European firms. *Industrial and Corporate Change*, 29(3), 855-875. https://doi.org/10.1093/icc/dtaa003

Comello, S., Glenk, G., & Reichelstein, S. (2021). Transitioning to clean energy transportation services: Life-cycle cost analysis for vehicle fleets. *Applied Energy*, 285, 116408. https://doi.org/10.1016/j.apenergy.2020.116408

Cox, B., Bauer, C., Mendoza Beltran, A., van Vuuren, D. P., & Mutel, C. L. (2020). Life cycle environmental and cost comparison of current and future passenger cars under different energy scenarios. *Applied Energy*, 269, 115021. https://doi.org/10.1016/j.apenergy.2020.115021

Dangelico, R. M., & Pujari, D. (2010). Mainstreaming green product innovation: Why and how companies integrate environmental sustainability. *Journal of Business Ethics*, 95(3), 471-486. https://doi.org/10.1007/s10551-010-0434-0

Davis, S., & Boundy, R. G. (2021). *Transportation Energy Data Book: Edition 39*. Oak Ridge, Tennessee: Oak Ridge National Laboratory (ORNL).

Dreier, D., Silveira, S., Khatiwada, D., Fonseca, K. V. O., Nieweglowski, R., & Schepanski, R. (2019). The influence of passenger load, driving cycle, fuel price and different types of buses on the cost of transport service in the BRT system in Curitiba, Brazil. *Transportation*, 46(6), 2195-2242. https://doi.org/10.1007/s11116-018-9925-0

EBRD. (2020). *Multilateral Development Banks' Climate Finance*. London: European Bank for Reconstruction and Development (EBRD).

Efficiency for Access Coalition (2019). *Use and benefits of solar water pumps*. Kenya, Tanzania and Uganda Consumer Research, CLASP, Washington, DC. Available online https://www.clasp.ngo/research/all/use-and-benefits-of-solar-water-pumps-2/

Efficiency for Access Coalition (2020). *Solar milling: Exploring market requirements to close the commercial viability gap*. CLASP, Washington, DC. Available online https://storage.googleapis.com/e4a-website-assets/SolarMilling_Market-Requirements.pdf

Falcão, E. A. M., Teixeira, A. C. R., & Sodré, J. R. (2017). Analysis of CO_2 emissions and techno-economic feasibility of an electric commercial vehicle. *Applied Energy*, 193, 297-307. https://doi.org/10.1016/j.apenergy.2017.02.050

Fama, E. F., & French, K. R. (2007). Disagreement, tastes, and asset prices. *Journal of Financial Economics*, 83(3), 667-689. https://doi.org/10.1016/j.jfineco.2006.01.003

Filippini, M., Geissmann, T., Karplus, V. J., & Zhang, D. (2020). The productivity impacts of energy efficiency programs in developing countries: Evidence from iron and steel firms in China. *China Economic Review*, *59*, 101364. https://doi.org/10.1016/j.chieco.2019.101364

Friedman, C., Becker, N., & Erell, E. (2014). Energy retrofit of residential building envelopes in Israel: A cost-benefit analysis. *Energy*, 77, 183-193. https://doi.org/https://doi.org/10.1016/j.energy.2014.06.019

FS, UNEP, & BNEF. (2020). *Global Trends in Renewable Energy Investment 2020*. Frankfurt School of Finance & Management (FS), United Nations Environmental Programme Collaborating Centre (UNEP), and BloombergNEF.

Gaddy, B. E., Sivaram, V., Jones, T. B., & Wayman, L. (2017). Venture Capital and Cleantech: The wrong model for energy innovation. *Energy Policy*, *102*, 385-395. https://doi.org/10.1016/j.enpol.2016.12.035

Gambhir, A., Tse, L. K. C., Tong, D., & Martinez-Botas, R. (2015). Reducing China's road transport sector CO_2 emissions to 2050: Technologies, costs and decomposition analysis. *Applied Energy*, 157, 905-917. https://doi.org/10.1016/j.apenergy.2015.01.018

Gong, R., Wu, Y. Q., Chen, F. W., & Yan, T. H. (2020). Labor costs, market environment and green technological innovation: Evidence from high-pollution firms. *International Journal* *of Environmental Research and Public Health*, *17*(2), 522. https://doi.org/10.3390/ijerph17020522

Haas, J., Moreno-Leiva, S., Junne, T., Chen, P.-J., Pamparana, G., Nowak, W., Kracht, W., & Ortiz, J. M. (2020). Copper mining: 100% solar electricity by 2030? *Applied Energy*, *262*, 114506. https://doi.org/10.1016/j.apenergy.2020.114506

Haider, S., & Bhat, J. A. (2020). Does total factor productivity affect the energy efficiency: Evidence from the Indian paper industry. *International Journal of Energy Sector Management*, *14*(1), 108-125. https://doi.org/10.1108/IJESM-11-2018-0010

Hamamoto, M. (2006). Environmental regulation and the productivity of Japanese manufacturing industries. *Resource and Energy Economics*, 28(4), 299-312. https://doi.org/10.1016/j.reseneeco.2005.11.001

Hamza, K., Laberteaux, K. P., & Chu, K.-C. (2021). On modeling the cost of ownership of plug-in vehicles. *World Electric Vehicle Journal*, 12(1). https://doi.org/10.3390/wevj12010039

Harchaoui, S., & Chatzimpiros, P. (2017). Reconstructing production efficiency, land use and trade for livestock systems in historical perspective. The case of France, 1961–2010. *Land Use Policy*, 67, 378-386. https://doi.org/https://doi.org/10.1016/j.landusepol.2017.05.028

Hasan, M. A., Frame, D. J., Chapman, R., & Archie, K. M. (2021). Costs and emissions: Comparing electric and petrol-powered cars in New Zealand. *Transportation Research PartD: Transport and Environment*, 90, 102671. https://doi.org/10.1016/j.trd.2020.102671

Hasanbeigi, A., Morrow, W. R., Masanet, E., Sathaye, J., & Xu, T. (2012). *Assessment of Energy Efficiency Improvement and CO2 Emission Reduction Potentials in the Cement Industry in China*. Berkeley: Lawrence Berkeley National Laboratory

Hassan, A. (2019). Do renewable energy incentive policies improve the performance of energy firms? Evidence from OECD countries. *OPEC Energy Review*, 43(2), 168-192. https://doi.org/10.1111/opec.12146

He, X., Wang, F., Wallington, T. J., Shen, W., Melaina, M. W., Kim, H. C., De Kleine, R., Lin, T., Zhang, S., Keoleian, G. A., Lu, X., & Wu, Y. (2021). Well-to-wheels emissions, costs, and feedstock potentials for light-duty hydrogen fuel cell vehicles in China in 2017 and 2030. *Renewable and Sustainable Energy Reviews*, 137, 110477. https://doi.org/10.1016/j.rser.2020.110477

Holland, S. P., Mansur, E. T., Muller, N. Z., & Yates, A. J. (2021). The environmental benefits of transportation electrification: Urban buses. *Energy Policy*, 148, 111921. https://doi.org/10.1016/j.enpol.2020.111921

Horváthová, E. (2012). The impact of environmental performance on firm performance: Short-term costs and long-term benefits? *Ecological Economics*, 84, 91-97. https://doi.org/10.1016/j.ecolecon.2012.10.001

Hottenrott, H., Rexhäuser, S., & Veugelers, R. (2016). Organisational change and the productivity effects of green technology adoption. *Resource and Energy Economics*, *43*, 172-194. https://doi.org/10.1016/j.reseneeco.2016.01.004

Hrovatin, N., Dolšak, N., & Zorić, J. (2016). Factors impacting investments in energy efficiency and clean technologies: empirical evidence from Slovenian manufacturing firms. *Journal of Cleaner Production*, *127*, 475-486. https://doi.org/10.1016/j.jclepro.2016.04.039

Huang, Y. H., & Wu, J. H. (2021). Bottom-up analysis of energy efficiency improvement and CO_2 emission reduction potentials in the cement industry for energy transition: An application of extended marginal abatement cost curves. *Journal of Cleaner Production*, *296*, 126619. https://doi.org/10.1016/j.jclepro.2021.126619

IEA (2019). *Energy efficiency 2019*. Paris: International Energy Agency (IEA). Available online https://iea.blob.core.windows.net/assets/8441ab46-9d86-47eb-b1fc-cb36fc3e7143/Energy_Efficiency_2019.pdf

IEA (2020a). *World Energy Investment 2020*. Paris: IEA.

IEA (2020b). *Energy efficiency 2020*. Paris: IEA.

IEA (2020c). *Energy efficiency indicators: Highlights*. Paris: IEA.

IEA (2021a). *Net Zero by 2050: A roadmap for the global energy system*. Paris: IEA.

IEA (2021b). *Data and statistics: Data tables*. Paris: IEA. Available online https://www.iea.org/data-and-statistics/data-tables

IFC (2019). *The Market Opportunity for Productive Use Leveraging Solar Energy (PULSE) in Sub-Saharan Africa*. Washington, DC: International Finance Corporation (IFC). Available online https://www.lightingglobal.org/wp-content/uploads/2019/09/PULSE-Report.pdf

Inoue, N., & Matsumoto, S. (2019). An examination of losses in energy savings after the Japanese Top Runner Program? *Energy Policy*, *124*, 312-319. https://doi.org/10.1016/j.enpol.2018.09.040

Jacobsen, G. D., Kotchen, M. J., & Vandenbergh, M. P. (2012). The behavioral response to voluntary provision of an environmental public good: Evidence from residential electricity demand. *European Economic Review*, 56(5), 946-960. https://doi.org/https://doi.org/10.1016/j.euroecorev.2012.02.008

Jaffe, A. B., & Palmer, K. (1997). Environmental regulation and innovation: A panel data study. *The Review of Economics and Statistics*, *79*(4), 610-619. https://www.jstor.org/stable/2951413

Kumar, S. (2016). Assessment of renewables for energy security and carbon mitigation in Southeast Asia: The case of Indonesia and Thailand. *Applied Energy*, 163, 63-70. https://doi.org/10.1016/j.apenergy.2015.11.019

Lanoie, P., Patry, M., & Lajeunesse, R. (2008). Environmental regulation and productivity: testing the porter hypothesis. *Journal of Productivity Analysis*, 30(2), 121-128. http://www.jstor.org.ezproxy.mnsu.edu/stable/41770359

Lebeau, K., Lebeau, P., Macharis, C., & Van Mierlo, J. (2013). *How expensive are electric vehicles? A total cost of ownership analysis*. 2013 World Electric Vehicle Symposium and Exhibition (EVS27).

Leoncini, R., Marzucchi, A., Montresor, S., Rentocchini, F., & Rizzo, U. (2019). 'Better late than never': the interplay between green technology and age for firm growth. *Small Business Economics*, *52*(4), 891-904. https://doi.org/10.1007/s11187-017-9939-6

Lévay, P. Z., Drossinos, Y., & Thiel, C. (2017). The effect of fiscal incentives on market penetration of electric vehicles: A pairwise comparison of total cost of ownership. *Energy Policy*, 105, 524-533. https://doi.org/10.1016/j.enpol.2017.02.054

Li, H., Zhou, H., Liu, K., Gao, X., & Li, X. (2021a). Retrofit application of traditional petroleum chemical technologies to coal chemical industry for sustainable energy-efficiency production. *Energy*, *218*, 119493. https://doi.org/10.1016/j.energy.2020.119493

Li, J., Liang, M., Cheng, W., & Wang, S. (2021b). Life cycle cost of conventional, battery electric, and fuel cell electric vehicles considering traffic and environmental policies in China.

International Journal of Hydrogen Energy, 46(14), 9553-9566. https://doi.org/10.1016/j.ijhydene.2020.12.100

Li, S.-h., Guo, L.-j., Cao, C.-g., & Li, C.-f. (2021c). Integrated assessment of carbon footprint, energy budget and net ecosystem economic efficiency from rice fields under different tillage modes in central China. *Journal of Cleaner Production*, 295, 126398. https://doi.org/https://doi.org/10.1016/j.jclepro.2021.126398

Li, Y., & Kimura, S. (2021). Economic competitiveness and environmental implications of hydrogen energy and fuel cell electric vehicles in ASEAN countries: The current and future scenarios. *Energy Policy*, 148, 111980. https://doi.org/10.1016/j.enpol.2020.111980

Liang, Y., Yu, B., & Wang, L. (2019). Costs and benefits of renewable energy development in China's power industry. *Renewable Energy*, 131, 700-712. https://doi.org/10.1016/j.renene.2018.07.079

Lio, M., & Liu, M. C. (2006). ICT and agricultural productivity: Evidence from cross-country data. *Agricultural Economics*, 34(3), 221-228. https://doi.org/10.1111/j.1574-0864.2006.00120.x

Liu, Y., Liu, T., Ye, S., & Liu, Y. (2018). Cost-benefit analysis for energy efficiency retrofit of existing buildings: A case study in China. *Journal of Cleaner Production*, 177, 493-506. https://doi.org/https://doi.org/10.1016/j.jclepro.2017.12.225

Lohwanitchai, K., & Jareemit, D. (2021). Modeling energy efficiency performance and cost-benefit analysis achieving net-zero energy building design: Case studies of three representative offices in Thailand. *Sustainability*, 13(9). https://doi.org/10.3390/su13095201

Luddeni, G., Krarti, M., Pernigotto, G., & Gasparella, A. (2018). An analysis methodology for large-scale deep energy retrofits of existing building stocks: Case study of the Italian office building. *Sustainable Cities and Society*, 41, 296-311. https://doi.org/https://doi.org/10.1016/j.scs.2018.05.038

Mai, T., Cole, W., & Reimers, A. (2019). Setting cost targets for zero-emission electricity generation technologies. *Applied Energy*, 250, 582-592. https://doi.org/10.1016/j.apenergy.2019.05.001

Majumder, S., De, K., Kumar, P., Sengupta, B., & Biswas, P. K. (2021). Techno-commercial analysis of sustainable E-bus-based public transit systems: An Indian case study. *Renewable and Sustainable Energy Reviews*, 144, 111033. https://doi.org/10.1016/j.rser.2021.111033

McAndrew, R., Mulcahy, R., Gordon, R., & Russell-Bennett, R. (2021). Household energy efficiency interventions: A systematic literature review. *Energy Policy*, 150, 112136. https://doi.org/https://doi.org/10.1016/j.enpol.2021.112136

Morrow III, W. R., Hasanbeigi, A., Sathaye, J., & Xu, T. (2013). *Assessment of Energy Efficiency Improvement and CO2 Emission Reduction Potentials in India's Cement Industry*. Berkeley: Lawrence Berkeley National Laboratory.

Mustapa, S. I., Ayodele, B. V., Mohamad Ishak, W. W., & Ayodele, F. O. (2020). Evaluation of cost competitiveness of electric vehicles in Malaysia using life cycle cost analysis approach. *Sustainability*, 12(13). https://doi.org/10.3390/su12135303

Nassif, G. G., & Almeida, S. C. A. D. (2020). Impact of powertrain hybridization on the performance and costs of a fuel cell electric vehicle. *International Journal of Hydrogen Energy*, 45(41), 21722-21737. https://doi.org/10.1016/j.ijhydene.2020.05.138

Ng, A., & Zheng, D. (2018). Let's agree to disagree! On payoffs and green tastes in green energy investments. *Energy Economics*, 69, 155-169. https://doi.org/10.1016/j.eneco.2017.10.023

Nikolidakis, S. A., Kandris, D., Vergados, D. D., & Douligeris, C. (2015). Energy efficient automated control of irrigation in agriculture by using wireless sensor networks. *Computers and Electronics in Agriculture*, 113, 154-163. https://doi.org/https://doi.org/10.1016/j.compag.2015.02.004

Nisar, S., Benbi, D. K., & Toor, A. S. (2021). Energy budgeting and carbon footprints of three tillage systems in maize-wheat sequence of north-western Indo-Gangetic Plains. *Energy*, 229, 120661. https://doi.org/https://doi.org/10.1016/j.energy.2021.120661

OECD (2001). *Measuring productivity - OECD manual*. Paris: OECD.

OECD (2005). *Oslo Manual*. Statistical Office of the European Communities. Luxembourg: OECD.

OECD and FAO (2018). *OECD-FAO Agricultural Outlook 2018-2027*, OECD Publishing, Paris and Food and Agriculture Organization (FAO) of the United Nations, Rome. https://doi.org/10.1787/agr_outlook-2018-en

Opoku, R., Edwin, I. A., & Agyarko, K. A. (2019). Energy efficiency and cost saving opportunities in public and commercial buildings in developing countries – The case of air-conditioners in Ghana. *Journal of Cleaner Production*, 230, 937-944. https://doi.org/https://doi.org/10.1016/j.jclepro.2019.05.067

Ouyang, D., Zhou, S., & Ou, X. (2021). The total cost of electric vehicle ownership: A consumer-oriented study of China's post-subsidy era. *Energy Policy*, 149, 112023. https://doi.org/10.1016/j.enpol.2020.112023

Palčič, I., & Prester, J. (2020). Impact of advanced manufacturing technologies on green innovation. *Sustainability (Switzerland)*, *12*(8), 3499. https://doi.org/10.3390/SU12083499

Palmer, K., Tate, J. E., Wadud, Z., & Nellthorp, J. (2018). Total cost of ownership and market share for hybrid and electric vehicles in the UK, US and Japan. *Applied Energy*, 209, 108-119. https://doi.org/10.1016/j.apenergy.2017.10.089

Palmer, M., & Truong, Y. (2017). The impact of technological green new product introductions on firm profitability. *Ecological Economics*, *136*, 86-93. https://doi.org/10.1016/j.ecolecon.2017.01.025

Pan, X., Ai, B., Li, C., Pan, X., & Yan, Y. (2019). Dynamic relationship among environmental regulation, technological innovation and energy efficiency based on large scale provincial panel data in China. *Technological Forecasting and Social Change*, *144*, 428-435. https://doi.org/10.1016/j.techfore.2017.12.012

Papadopoulos, S., Kontokosta, C. E., Vlachokostas, A., & Azar, E. (2019). Rethinking HVAC temperature setpoints in commercial buildings: The potential for zero-cost energy savings and comfort improvement in different climates. *Building and Environment*, 155, 350-359. https://doi.org/https://doi.org/10.1016/j.buildenv.2019.03.062

Park, W. Y., Shah, N., & Phadke, A. (2019). Enabling access to household refrigeration services through cost reductions from energy efficiency improvements. *Energy Efficiency*, 12(7), 1795-1819. https://doi.org/10.1007/s12053-019-09807-w

Peng, J., Xie, R., Ma, C., & Fu, Y. (2021). Market-based environmental regulation and total factor productivity: Evidence from Chinese enterprises. *Economic Modelling*, 95, 394-407. https://doi.org/10.1016/j.econmod.2020.03.006

Popp, D., Hascic, I., & Medhi, N. (2011). Technology and the diffusion of renewable energy. *Energy Economics*, 33(4), 648-662. https://doi.org/10.1016/j.eneco.2010.08.007

Porter, M. E. (1991). America's green strategy. *Scientific American*, 264 (4), 168.

Porter, M. E., & van der Linde, C. (1995). Toward a new conception of the environment-competitiveness relationship. *The Journal of Economic Perspectives*, 9(4), 97-118. http://www.jstor.org/stable/2138392

Przychodzen, W., Gómez-Bezares, F., & Przychodzen, J. (2018). Green information technologies practices and financial performance – The empirical evidence from German publicly traded companies. *Journal of Cleaner Production*, 201, 570-579. https://doi.org/10.1016/j.jclepro.2018.08.081

Qi, J., Tang, X., & Xi, X. (2021). The Size Distribution of Firms and Industrial Water Pollution: A Quantitative Analysis of China. *American Economic Journal: Macroeconomics*, *13*(1), 151-183. https://doi.org/10.1257/mac.20180227

Rakhshan, K., & Friess, W. A. (2017). Effectiveness and viability of residential building energy retrofits in Dubai. *Journal of Building Engineering*, 13, 116-126. https://doi.org/https://doi.org/10.1016/j.jobe.2017.07.010

Rath, B. N., Akram, V., Bal, D. P., & Mahalik, M. K. (2019). Do fossil fuel and renewable energy consumption affect total factor productivity growth? Evidence from cross-country data with policy insights. *Energy Policy*, *127*, 186-199. https://doi.org/10.1016/j.enpol.2018.12.014

Ren, H., Tibbs, M., McLauchlan, C., Ma, Z., & Harrington, L. (2021). Refrigerator cost trap for low-income households: Developments in measurement and verification of appliance replacements. *Energy for Sustainable Development*, 60, 1-14. https://doi.org/https://doi.org/10.1016/j.esd.2020.11.001

Roberts, M. B., Bruce, A., & MacGill, I. (2019). Impact of shared battery energy storage systems on photovoltaic self-consumption and electricity bills in apartment buildings. *Applied Energy*, 245, 78-95. https://doi.org/https://doi.org/10.1016/j.apenergy.2019.04.001

Rubashkina, Y., Galeotti, M., & Verdolini, E. (2015). Environmental regulation and competitiveness: Empirical evidence on the Porter Hypothesis from European manufacturing sectors. *Energy Policy*, 83, 288-300. https://doi.org/10.1016/j.enpol.2015.02.014

Ruffini, E., & Wei, M. (2018). Future costs of fuel cell electric vehicles in California using a learning rate approach. *Energy*, 150, 329-341. https://doi.org/10.1016/j.energy.2018.02.071

Saha, S., Nielsen, I., & Moon, I. (2017). Optimal retailer investments in green operations and preservation technology for deteriorating items. *Journal of Cleaner Production*, 140, 1514-1527. https://doi.org/10.1016/j.jclepro.2016.09.229

Schulte, R. H., & Fletcher, F. C. (2021). Why individual electric utilities cannot achieve 100 % clean energy. *The Electricity Journal*, 34(2), 106909. https://doi.org/10.1016/j.tej.2020.106909

Shapiro, J. S., & Walker, R. (2018). Why Is Pollution from US Manufacturing Declining? The Roles of Environmental Regulation, Productivity, and Trade. *American Economic Review*, *108*(12), 3814-3854. https://doi.org/10.1257/aer.20151272

Simmons, R. A., Shaver, G. M., Tyner, W. E., & Garimella, S. V. (2015). A benefit-cost assessment of new vehicle technologies and fuel economy in the U.S. market. *Applied Energy*, 157, 940-952. https://doi.org/10.1016/j.apenergy.2015.01.068

Sohag, K., Chukavina, K., & Samargandi, N. (2021). Renewable energy and total factor productivity in OECD member countries. *Journal of Cleaner Production*, *296*, 126499. https://doi.org/10.1016/j.jclepro.2021.126499

Steckel, J. C., & Jakob, M. (2018). The role of financing cost and de-risking strategies for clean energy investment. *International Economics*, 155, 19-28. https://doi.org/10.1016/j.inteco.2018.02.003

Stucki, T. (2019). Which firms benefit from investments in green energy technologies? – The effect of energy costs. *Research Policy*, *48*(3), 546-555. https://doi.org/10.1016/j.respol.2018.09.010

Suástegui Macías, J. A., Pérez Tello, C., Acuña Ramírez, A., Lambert Arista, A. A., Magaña Almaguer, H. D., Rosales Escobedo, P. F., & Ruelas Puente, A. H. (2018). Assessment of electrical saving from energy efficiency programs in the residential sector in Mexicali, Mexico. *Sustainable Cities and Society*, 38, 795-805. https://doi.org/https://doi.org/10.1016/j.scs.2018.01.031

Tarei, P. K., Chand, P., & Gupta, H. (2021). Barriers to the adoption of electric vehicles: Evidence from India. *Journal of Cleaner Production*, 291, 125847. https://doi.org/10.1016/j.jclepro.2021.125847

Teoh, L. E., Khoo, H. L., Goh, S. Y., & Chong, L. M. (2018). Scenario-based electric bus operation: A case study of Putrajaya, Malaysia. *International Journal of Transportation Science and Technology*, 7(1), 10-25. https://doi.org/10.1016/j.ijtst.2017.09.002

Thompson, S. T., James, B. D., Huya-Kouadio, J. M., Houchins, C., DeSantis, D. A., Ahluwalia, R., Wilson, A. R., Kleen, G., & Papageorgopoulos, D. (2018). Direct hydrogen fuel cell electric vehicle cost analysis: System and high-volume manufacturing description, validation, and outlook. *Journal of Power Sources*, 399, 304-313. https://doi.org/10.1016/j.jpowsour.2018.07.100

Thorne, R. J., Hovi, I. B., Figenbaum, E., Pinchasik, D. R., Amundsen, A. H., & Hagman, R. (2021). Facilitating adoption of electric buses through policy: Learnings from a trial in Norway. *Energy Policy*, 155, 112310. https://doi.org/10.1016/j.enpol.2021.112310

Tiefenbeck, V., Staake, T., Roth, K., & Sachs, O. (2013). For better or for worse? Empirical evidence of moral licensing in a behavioral energy conservation campaign. *Energy Policy*, 57, 160-171. https://doi.org/https://doi.org/10.1016/j.enpol.2013.01.021

Timilsina, G.R. (2020). *Demystifying the Costs of Electricity Generation Technologies*. World Bank Policy Research Working Paper, WPS9303. Washington, DC: World Bank.

Timilsina, G.R. (2021). *Economics of Distributed Photovoltaics: An Illustration from Bangladesh*. World Bank Policy Research Working Paper, WPS9699. Washington, DC: World Bank.

Tugcu, C. T., & Tiwari, A. K. (2016). Does renewable and/or non-renewable energy consumption matter for total factor productivity (TFP) growth? Evidence from the BRICS. *Renewable and Sustainable Energy Reviews*, *65*, 610-616. https://doi.org/10.1016/j.rser.2016.07.016

Ugur, M., Trushin, E., Solomon, E., & Guidi, F. (2016). R&D and productivity in OECD firms and industries: A hierarchical meta-regression analysis. *Research Policy*, 45(10), 2069-2086. https://doi.org/10.1016/j.respol.2016.08.001

Unver, U., & Kara, O. (2019). Energy efficiency by determining the production process with the lowest energy consumption in a steel forging facility. *Journal of Cleaner Production*, *215*, 1362-1370. https://doi.org/10.1016/j.jclepro.2019.01.168

van Leeuwen, G., & Mohnen, P. (2017). Revisiting the Porter hypothesis: an empirical analysis of Green innovation for the Netherlands. *Economics of Innovation and New Technology*, 26(1-2), 63-77. https://doi.org/10.1080/10438599.2016.1202521

van Velzen, A., Annema, J. A., van de Kaa, G., & van Wee, B. (2019). Proposing a more comprehensive future total cost of ownership estimation framework for electric vehicles. *Energy Policy*, 129, 1034-1046. https://doi.org/10.1016/j.enpol.2019.02.071

Vellini, M., Bellocchi, S., Gambini, M., Manno, M., & Stilo, T. (2020). Impact and costs of proposed scenarios for power sector decarbonisation: An Italian case study. *Journal of Cleaner Production*, 274, 123667. https://doi.org/10.1016/j.jclepro.2020.123667

Vora, A. P., Jin, X., Hoshing, V., Saha, T., Shaver, G., Varigonda, S., Wasynczuk, O., & Tyner, W. E. (2017). Design-space exploration of series plug-in hybrid electric vehicles for medium-duty truck applications in a total cost-of-ownership framework. *Applied Energy*, 202, 662-672. https://doi.org/10.1016/j.apenergy.2017.05.090

Wan, D., Xue, R., Linnenluecke, M., Tian, J., & Shan, Y. (2021). The impact of investor attention during COVID-19 on investment in clean energy versus fossil fuel firms. *Finance Research Letters*, 101955. https://doi.org/10.1016/j.frl.2021.101955

Weiss, M., Zerfass, A., & Helmers, E. (2019). Fully electric and plug-in hybrid cars - An analysis of learning rates, user costs, and costs for mitigating CO_2 and air pollutant emissions. *Journal of Cleaner Production*, 212, 1478-1489. https://doi.org/10.1016/j.jclepro.2018.12.019

Whiston, M. M., Lima Azevedo, I. M., Litster, S., Samaras, C., Whitefoot, K. S., & Whitacre, J. F. (2021). Hydrogen storage for fuel cell electric vehicles: Expert elicitation and a levelized cost of driving model. *Environmental Science and Technology*, 55(1), 553-562. https://doi.org/10.1021/acs.est.0c04145

Witt, S. M., Stults, S., Rieves, E., Emerson, K., & Mendoza, D. L. (2019). Findings from a pilot light-emitting diode (LED) bulb exchange program at a neighborhood scale. *Sustainability*, 11(14). https://doi.org/10.3390/su11143965

Wurlod, J. D., & Noailly, J. (2018). The impact of green innovation on energy intensity: An empirical analysis for 14 industrial sectors in OECD countries. *Energy Economics*, *71*, 47-61. https://doi.org/10.1016/j.eneco.2017.12.012

Würzburg, K., Labandeira, X., & Linares, P. (2013). Renewable generation and electricity prices: Taking stock and new evidence for Germany and Austria. *Energy Economics*, 40, S159-S171. https://doi.org/10.1016/j.eneco.2013.09.011

Xylia, M., Silveira, S., & Morfeldt, J. (2017). Implications of an energy efficiency obligation scheme for the Swedish energy-intensive industries: an evaluation of costs and benefits. *Energy Efficiency*, *10*(1), 151-169. https://doi.org/10.1007/s12053-016-9446-7

Zhang, D., & Vigne, S. A. (2021). How does innovation efficiency contribute to green productivity? A financial constraint perspective. *Journal of Cleaner Production*, *280*, 124000. https://doi.org/10.1016/j.jclepro.2020.124000

Zhang, D., Du, W., Zhuge, L., Tong, Z., & Freeman, R. B. (2019). Do financial constraints curb firms' efforts to control pollution? Evidence from Chinese manufacturing firms. *Journal of Cleaner Production*, *215*, 1052-1058. https://doi.org/10.1016/j.jclepro.2019.01.112

Zhang, D., Zhuge, L., & Freeman, R. B. (2020). Firm dynamics of hi-tech start-ups: Does innovation matter? *China Economic Review*, *59*, 101370. https://doi.org/10.1016/j.chieco.2019.101370

Zheng, X., Wu, C., & He, S. (2021). Impacts of China's differential electricity pricing on the productivity of energy-intensive industries. *Energy Economics*, *94*, 105050. https://doi.org/10.1016/j.eneco.2020.105050

Zhou, Z., Zhang, S., Wang, C., Zuo, J., He, Q., & Rameezdeen, R. (2016). Achieving energy efficient buildings via retrofitting of existing buildings: a case study. *Journal of Cleaner Production*, 112, 3605-3615. https://doi.org/https://doi.org/10.1016/j.jclepro.2015.09.046

Zhu, X., Chen, Y., & Feng, C. (2018). Green total factor productivity of China's mining and quarrying industry: A global data envelopment analysis. *Resources Policy*, *57*, 1-9. https://doi.org/10.1016/j.resourpol.2017.12.009

Zuberi, M. J. S., & Patel, M. K. (2019). Cost-effectiveness analysis of energy efficiency measures in the Swiss chemical and pharmaceutical industry. *International Journal of Energy Research*, *43*(1), 313-336. https://doi.org/10.1002/er.4267